RV College of Engineering

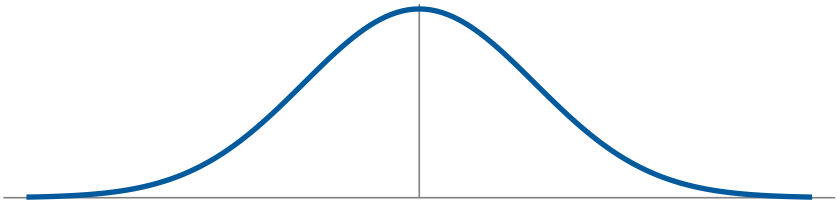
Department of Industrial Engineering and Management

Statistics for Data Analytics

Course Code: IM241AT

Semester IV • Professional Core Course

Unit 2



A Student Compilation for Study and Reference

Semester: IV						
STATISTICS FOR DATA ANALYTICS **Category: Professional Core Course** **(Theory)**						
Course Code	**:**	**IM241AT**		**CIE**	**:**	**100 Marks**
Credits: L:T:P	**:**	**3:0:0**		**SEE**	**:**	**100 Marks**
Total Hours	**:**	**45L**		**SEE Duration**	**:**	**3.00 Hours**

Unit-I	**09 Hrs**
Data Summary and Presentation: Data types, tabular and graphical displays: Stem and Leaf diagrams, Histograms, Box plots, Radar diagrams. **Concepts of Probability:** Sample spaces and Events, Interpretations of probability, Addition rules, Conditional probability, Multiplication and Total probability rules, Independence, Bayes Theorem. Numerical Problems	
Unit-II	**09 Hrs**
Concepts of Random Variables: Random Variables, Discrete and continuous random variables. Probability distributions and mass functions, Numerical Problems **Discrete Probability Distributions:** Discrete uniform, Binominal, Poisson, Geometric, Negative binomial, Applications, Numerical Problems.	
Unit-III	**09 Hrs**
Continuous Probability Distributions: Continuous uniform, Normal, Normal approximations, Exponential, Applications, Numerical Problems. **Sampling Distribution:** Sampling distribution, Central Limit Theorem, Sampling distribution of means and derived quantities, Numerical Problems.	
Unit-IV	**09 Hrs**
Simple Linear Regression and Correlation: Empirical models, Simple Linear Regression, Properties of Least square Estimators and Estimation of variances, Common abuses of regression, Prediction of new observations, Correlation, Numerical Problems. Interpretation of graphical output from software packages such as Minitab	
Unit-V	**09 Hrs**
Statistical Inference for a single sample: Hypothesis testing, Inference on the mean of a normal population (variance known and unknown), Inference on the variance of a Normal population, Testing for Goodness of Fit, Tests of association, Numerical Problems **Interval Estimation:** Confidence Intervals on mean (variance known and unknown), and variance of a normal population	

Course Outcomes: After completing the course, the students will be able to	
CO1	Describe and report data set using data analysis, presentation and interpretation techniques to understand various phenomena in the fields of science and engineering.
CO2	Apply various statistical processing techniques to handle a set of data to estimate probabilities.
CO3	Apply an appropriate statistical tool and analyze a specific set of data to estimate and draw conclusions about population parameters
CO4	Draw inferences about population parameters and relations between variables based on analysis of sample data

References Books	
1.	Engineering Statistics, Douglas C. Montgomery, George C. Runger, Norma FarisHubele, 5th Edition, 2011, John Wiley & Sons, Inc., ISBN-13: 978- 0-470-63147-8
2.	Applied statistics and Probability for Engineers, Douglas C Montgomery, George C Runger, Wiley, 4th Edition, 2007, Asia Student Edition, ISBN: 978-81-265-2315-3.
3.	Statistics for Management, Richard I Levin, David S Rubin, 7th Edition, 1997, Prentice Hall India, ISBN: 9780134762920.
4.	Probability and Statistics for Engineers and Scientists, Walpole, Myers, Myers, Ye, 8th Edition, 2007, Pearson Education Inc., ISBN: 978-81-317-1552-9.
5.	Softwares : Microsoft Excel / Minitab / Matlab / R

6.	Online resources: a) http://172.16.44.44/nnptel.html - choose NOC:Introduction to Data Analytics(Course sponsored by Aricent) b) https://onlinecourses.nptel.ac.in/noc22_mg31/preview -Introduction to probability and Statistics c) https://newonlinecourses.science.psu.edu/statprogram/undergraduate-studies d) https://www.khanacademy.org/math/statistics-probability

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (THEORY)		
#	**COMPONENTS**	**MARKS**
1.	**QUIZZES:** Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted & Each Quiz will be evaluated for 10 Marks. **THE SUM OF TWO QUIZZES WILL BE THE FINAL QUIZ MARKS.**	**20**
2.	**TESTS:** Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. **FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.**	**40**
3.	**EXPERIENTIAL LEARNING:** Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20) **ADDING UPTO 40**	**40**
MAXIMUM MARKS FOR THE CIE THEORY		**100**

RUBRIC FOR SEMESTER END EXAMINATION (THEORY)		
Q. NO.	**CONTENTS**	**MARKS**
	PART A	
1	Objective type questions covering entire syllabus	20
	PART B (Maximum of TWO Sub-divisions only)	
2	Unit 1: (Compulsory)	16
3 & 4	Unit 2: (Internal Choice)	16
5 & 6	Unit 3: (Internal Choice)	16
7 & 8	Unit 4: (Internal Choice)	16
9 & 10	Unit 5: (Internal Choice)	16
	TOTAL	**100**

A Note on This Compilation

This book is **not** the official RV College of Engineering notes. It is a compilation of student notes together with the handwritten notes provided by **Dr. Ramaa A Nadig**, Department of Industrial Engineering and Management.

All information contained in this book has been drawn from her notes and lectures. The material has been organised, typeset, and arranged to follow the prescribed course syllabus, but it should be treated as a supplementary study aid rather than an official course document.

Every effort has been made to reproduce the source notes faithfully and to preserve their content without addition or deletion. Where the original handwritten working was unclear, the final answers recorded in the notes were treated as authoritative. Readers should always cross-check against the prescribed textbooks and their instructor's guidance.

Compiled from the notes and lectures of Dr. Ramaa A Nadig, Department of Industrial Engineering and Management.

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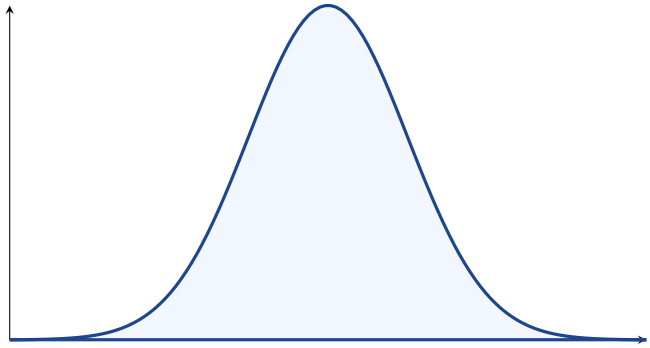
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Unit 2



Preface

This book is a polished, textbook-format rendering of a set of handwritten course notes on probability distributions. The material has been reorganised into logical chapters with connective prose, while every definition, formula, worked example, and final numerical answer from the original notes has been preserved.

The notes were intended to be reproduced one-to-one. Wherever the source text was legible, it is transcribed faithfully; only where the handwriting could not be read, or an intermediate algebraic step was omitted, has content been reconstructed from the surrounding context so that each derivation reaches the final answer recorded in the original. The final numerical answers in the original notes are treated as authoritative throughout. A short, honest list of every place where reconstruction or interpretation was required appears in Appendix A.

The book covers discrete models (geometric, negative binomial, hypergeometric, Poisson) and continuous models (general probability density functions, the continuous uniform, normal, exponential, Erlang, gamma, and Weibull distributions), together with the normal approximation to the binomial and Poisson distributions.

Chapter 1

Geometric and Negative Binomial Distributions

The binomial distribution counts the number of successes in a *fixed* number of Bernoulli trials. In this chapter we turn the question around: instead of fixing the number of trials, we keep performing independent Bernoulli trials (each with constant success probability p) until a prescribed number of successes has occurred, and we treat the *number of trials* as the random variable. This leads first to the geometric distribution (wait for the first success) and then to its generalisation, the negative binomial distribution (wait for the rth success).

1.1 The Geometric Distribution

Again consider a series of Bernoulli trials — independent trials with constant probability p of a success on each trial. However, instead of a fixed number of trials, trials are conducted until a success is obtained. Let the random variable X denote the number of trials until the first success.

Definition 1.1: Geometric random variable

In a series of Bernoulli trials (independent trials with constant probability p of a success), let the random variable X denote the number of trials until the first success. Then X is a *geometric* random variable with parameter $0 < p < 1$ and

$$f(x) = (1-p)^{x-1}p, \qquad x = 1, 2, 3, \ldots$$

Key Formula

For a geometric random variable with parameter p,

$$f(x) = (1-p)^{x-1}p, \qquad \mu = \mathbb{E}(X) = \frac{1}{p}, \qquad \sigma^2 = \mathrm{Var}(X) = \frac{1-p}{p^2}.$$

Lack of Memory Property

A geometric random variable has been defined as the number of trials until the first success. However, because the trials are independent, the count of the number of trials until the next success can be started at any trial without changing the probability distribution of the random variable.

For example, in the transmission of bits, suppose that 100 bits are transmitted. The probability that the initial error occurs on bit 106 is the probability that the next six outcomes are $00000E$.

This probability is

$$(0.9)^5(0.1) = 0.059,$$

which is identical to the probability that the initial error occurs on bit 6.

The implication of using a geometric model is that the system presumably will not wear out: the probability of an error remains constant for all transmissions. In this sense, the geometric distribution is said to lack any memory.

1.2 The Negative Binomial Distribution

A generalisation of a geometric distribution, in which the random variable is the number of Bernoulli trials required to obtain r successes, results in the negative binomial distribution.

Definition 1.2: Negative binomial random variable

In a series of Bernoulli trials (independent trials with constant probability p of a success), let the random variable X denote the number of trials until r successes occur. Then X is a *negative binomial* random variable with parameters $0 < p < 1$ and $r = 1, 2, 3, \ldots$, and

$$f(x) = \binom{x-1}{r-1}(1-p)^{x-r}p^r, \qquad x = r, r+1, r+2, \ldots$$

Key Formula

For a negative binomial random variable with parameters p and r,

$$f(x) = \binom{x-1}{r-1}(1-p)^{x-r}p^r, \qquad \mathbb{E}(X) = \frac{r}{p}, \qquad \mathrm{Var}(X) = \frac{r(1-p)}{p^2}.$$

Because at least r trials are required to obtain r successes, the range of X is from r to ∞. In the special case that $r = 1$, a negative binomial random variable is a geometric random variable.

A Negative Binomial Variable as a Sum of Geometric Variables

The lack-of-memory property of a geometric random variable leads to the following interpretation. Let X denote the total number of trials required to obtain r successes. Let X_1 denote the number of trials required to obtain the first success, X_2 the number of extra trials required to obtain the second success, X_3 the number of extra trials to obtain the third success, and so forth. Then the total number of trials required to obtain r successes is

$$X = X_1 + X_2 + \cdots + X_r.$$

A negative binomial random variable can therefore be interpreted as the sum of r geometric random variables.

In a negative binomial experiment the number of successes is predetermined and the number of trials is random. Hence a negative binomial random variable can be considered the opposite of a binomial random variable: a binomial random variable is a count of the number of successes in n trials.

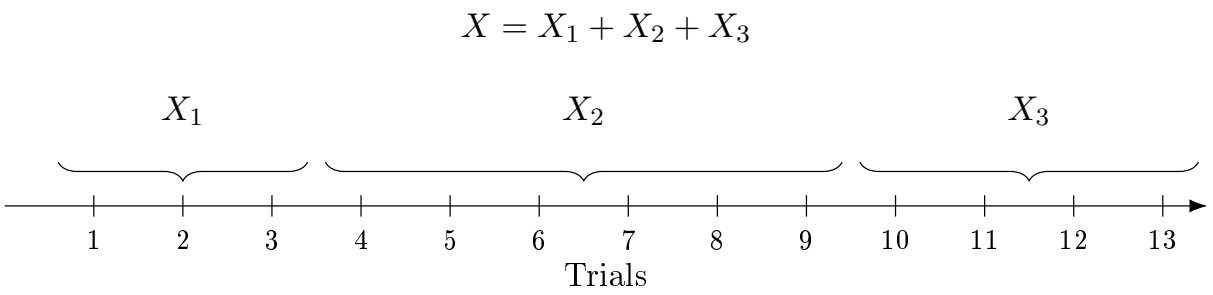


Figure 1.1: A negative binomial random variable ($r = 3$) represented as the sum of three independent geometric random variables.

1.3 Worked Problems

Example 1.1 (Geometric distribution with a given mean)**.** Suppose the random variable X has a geometric distribution with a mean of 2.5. Determine the following probabilities: (a) $\mathrm{P}(X = 1)$, (b) $\mathrm{P}(X = 4)$, (c) $\mathrm{P}(X = 5)$, (d) $\mathrm{P}(X \leq 3)$, (e) $\mathrm{P}(X > 3)$.

Solution. Since $\mathbb{E}(X) = 2.5 = \dfrac{1}{p}$, we have $p = 0.4$.

(a) $\mathrm{P}(X = 1) = (1 - 0.4)^0(0.4) = 0.4.$

(b) $\mathrm{P}(X = 4) = (1 - 0.4)^3(0.4) = 0.0864.$

(c) $\mathrm{P}(X = 5) = (1 - 0.4)^4(0.4) = 0.05184.$

(d) $$\begin{aligned}\mathrm{P}(X \leq 3) &= \mathrm{P}(X = 1) + \mathrm{P}(X = 2) + \mathrm{P}(X = 3)\\ &= (1 - 0.4)^0(0.4) + (1 - 0.4)^1(0.4) + (1 - 0.4)^2(0.4)\\ &= 0.7840.\end{aligned}$$

(e) $\mathrm{P}(X > 3) = 1 - \mathrm{P}(X \leq 3) = 1 - 0.7840 = 0.2160.$

Example 1.2 (Optical alignment)**.** The probability of a successful optical alignment in the assembly of an optical data storage product is 0.8. Assume the trials are independent. (a) What is the probability that the first successful alignment requires exactly four trials? (b) at most four trials? (c) at least four trials?

Solution. Let X denote the number of trials to obtain the first successful alignment. Then X is geometric with $p = 0.8$.

(a) $\mathrm{P}(X = 4) = (1 - 0.8)^3(0.8) = (0.2)^3(0.8) = 0.0064.$

(b) $$\begin{aligned}\mathrm{P}(X \leq 4) &= \mathrm{P}(X = 1) + \mathrm{P}(X = 2) + \mathrm{P}(X = 3) + \mathrm{P}(X = 4)\\ &= (0.2)^0(0.8) + (0.2)^1(0.8) + (0.2)^2(0.8) + (0.2)^3(0.8)\\ &= 0.8 + 0.16 + 0.032 + 0.0064 = 0.9984.\end{aligned}$$

(c) $$\begin{aligned}\mathrm{P}(X \geq 4) &= 1 - \mathrm{P}(X \leq 3) = 1 - \left[0.8 + 0.2(0.8) + (0.2)^2(0.8)\right]\\ &= 1 - 0.992 = 0.008.\end{aligned}$$

Example 1.3 (Carrier gene detection)**.** In a clinical study, volunteers are tested for a gene that has been found to increase the risk for a disease. The probability that a person carries the gene is 0.1. (a) What is the probability that four or more people will have to be tested before two

with the gene are detected? (b) How many people are expected to be tested before two with the gene are detected?

Solution. Let X denote the number of people tested to detect two who carry the gene. Then X is a negative binomial random variable with $p = 0.1$ and $r = 2$.

$$\begin{aligned}
\text{(a)} \quad \mathrm{P}(X \geq 4) &= 1 - \mathrm{P}(X < 4) = 1 - \big[\mathrm{P}(X = 2) + \mathrm{P}(X = 3)\big] \\
&= 1 - \left[\binom{1}{1}(1-0.1)^0(0.1)^2 + \binom{2}{1}(1-0.1)^1(0.1)^2\right] \\
&= 1 - (0.01 + 0.018) = 0.972. \\
\text{(b)} \quad \mathbb{E}(X) &= \frac{r}{p} = \frac{2}{0.1} = 20.
\end{aligned}$$

Example 1.4 (Connecting to a radio station)**.** Assume that each of your calls to a popular radio station has a probability 0.02 of connecting, that is, of not obtaining a busy signal. Assume that your calls are independent. (a) What is the probability that your first call that connects is your tenth call? (b) What is the probability that it requires more than five calls for you to connect? (c) What is the mean number of calls needed to connect?

Solution. Let X denote the number of calls needed to obtain a connection. Then X is geometric with $p = 0.02$.

$$\begin{aligned}
\text{(a)} \quad \mathrm{P}(X = 10) &= (1-0.02)^9(0.02) = (0.98)^9(0.02) = 0.0167. \\
\text{(b)} \quad \mathrm{P}(X > 5) &= 1 - \mathrm{P}(X \leq 5) \\
&= 1 - \big[\mathrm{P}(X = 1) + \mathrm{P}(X = 2) + \mathrm{P}(X = 3) + \mathrm{P}(X = 4) + \mathrm{P}(X = 5)\big] \\
&= 1 - \big[0.02 + 0.98(0.02) + (0.98)^2(0.02) \\
&\qquad + (0.98)^3(0.02) + (0.98)^4(0.02)\big] \\
&= 1 - 0.0961 = 0.9039.
\end{aligned}$$

This may also be found from "no connection in 5 trials" (a negative binomial / binomial argument):

$$\mathrm{P}(X > 5) = \binom{5}{0}(0.02)^0(0.98)^5 = 0.9039.$$

$$\text{(c)} \quad \mathbb{E}(X) = \frac{1}{0.02} = 50.$$

Example 1.5 (Fault-tolerant system)**.** A fault-tolerant system that processes transactions for a financial services firm uses three separate computers. If the operating computer fails, one of the two spares can be switched immediately online. After a second computer fails, the last computer can be immediately switched online. Assume that the probability of a failure during any transaction is 10^{-8} and that the transactions can be considered independent events. (a) What is the mean number of transactions before all computers have failed? (b) What is the variance of the number of transactions before all computers have failed?

Solution. Let X denote the number of transactions until all computers have failed. Then X is a negative binomial random variable with $p = 10^{-8}$ and $r = 3$.

$$\text{(a)} \quad \mathbb{E}(X) = \frac{r}{p} = \frac{3}{10^{-8}} = 3 \times 10^8.$$

$$\text{(b)} \quad \mathrm{Var}(X) = \frac{r(1-p)}{p^2} = \frac{3\,[\,1 - 10^{-8}\,]}{(10^{-8})^2} \approx 3 \times 10^{16}.$$

Chapter 2

The Hypergeometric Distribution

The binomial distribution applies to sampling *with* replacement (or, equivalently, to independent trials with constant success probability). When a finite collection of objects is sampled *without* replacement, the trials are no longer independent, and the hypergeometric distribution is required.

2.1 Definition and Properties

Consider a set of N objects, of which K are classified as successes and $N-K$ as failures. A sample of size n objects is selected randomly (without replacement) from the N objects, where $K \leq N$ and $n \leq N$.

Definition 2.1: Hypergeometric random variable

Let the random variable X denote the number of successes in the sample. Then X is a *hypergeometric* random variable and

$$f(x) = \frac{\binom{K}{x}\binom{N-K}{n-x}}{\binom{N}{n}}, \qquad x = \max\{0,\, n+K-N\} \text{ to } \min\{K, n\}.$$

The expression $\min\{K, n\}$ is used in the range of X because the maximum number of successes that can occur in the sample is the smaller of the sample size n and the number of successes available, K.

Key Formula

If X is a hypergeometric random variable with parameters N, K, and n, then

$$\mu = \mathbb{E}(X) = np, \qquad \sigma^2 = \mathrm{Var}(X) = np(1-p)\left(\frac{N-n}{N-1}\right), \qquad \text{where } p = \frac{K}{N}.$$

Here p is interpreted as the proportion of successes in the set of N objects.

The term $\dfrac{N-n}{N-1}$ is called the *finite population correction factor*.

2.2 Worked Problems

Example 2.1 (Unacceptable washers)**.** A lot of 75 washers contains 5 in which the variability in thickness around the circumference of the washer is unacceptable. A sample of 10 washers is selected at random without replacement. (a) What is the probability that none of the unacceptable washers is in the sample? (b) the probability that at least one unacceptable washer is in the sample? (c) the probability that exactly one unacceptable washer is in the sample? (d) the mean number of unacceptable washers in the sample?

Solution. Let X denote the number of unacceptable washers in the sample of 10.

$$\begin{aligned}
\text{(a)}\quad \mathrm{P}(X=0) &= \frac{\binom{5}{0}\binom{70}{10}}{\binom{75}{10}} = \frac{70!\,/\,(60!\,10!)}{75!\,/\,(65!\,10!)} \\
&= \frac{65\cdot 64\cdot 63\cdot 62\cdot 61}{75\cdot 74\cdot 73\cdot 72\cdot 71} = 0.4786. \\
\text{(b)}\quad \mathrm{P}(X\geq 1) &= 1-\mathrm{P}(X=0) = 1-0.4786 = 0.5214. \\
\text{(c)}\quad \mathrm{P}(X=1) &= \frac{\binom{5}{1}\binom{70}{9}}{\binom{75}{10}} = \frac{5\cdot 10\cdot 65\cdot 64\cdot 63\cdot 62}{75\cdot 74\cdot 73\cdot 72\cdot 71} = 0.3923. \\
\text{(d)}\quad \mathbb{E}(X) &= np = 10\left(\frac{5}{75}\right) = \frac{2}{3}.
\end{aligned}$$

Example 2.2 (Printed circuit cards)**.** Printed circuit cards are placed in a final test after being populated with semiconductor chips. A lot contains 140 cards, and 20 cards are selected without replacement for final testing. (a) If 20 cards are defective, what is the probability that at least one defective card is in the sample? (b) If 5 cards are defective, what is the probability that at least one defective card appears in the sample?

Solution. Let X denote the number of defective cards in the sample.

(a) Here $K=20$, $n=20$, $N=140$.

$$\begin{aligned}
\mathrm{P}(X=0) &= \frac{\binom{20}{0}\binom{120}{20}}{\binom{140}{20}} = \frac{120!\,/\,(100!\,20!)}{140!\,/\,(120!\,20!)} = 0.0356, \\
\mathrm{P}(X\geq 1) &= 1-\mathrm{P}(X=0) = 1-0.0356 = 0.9644.
\end{aligned}$$

(b) Here $K=5$, $n=20$, $N=140$.

$$\begin{aligned}
\mathrm{P}(X=0) &= \frac{\binom{5}{0}\binom{135}{20}}{\binom{140}{20}} = \frac{135!\,/\,(115!\,20!)}{140!\,/\,(120!\,20!)} = 0.4571, \\
\mathrm{P}(X\geq 1) &= 1-0.4571 = 0.5429.
\end{aligned}$$

Example 2.3 (Slitter assembly blades)**.** Magnetic tape is slit into $\frac{1}{2}''$ widths that are wound into cartridges. A slitter assembly contains 48 blades. Five blades are selected at random and evaluated each day for sharpness. If any dull blade is found, the assembly is replaced with a newly sharpened set of blades. (a) If 10 of the 48 blades in the assembly are dull, what is the probability that the assembly is replaced the first day it is evaluated? (b) If 10 of the blades in an assembly are dull, what is the probability that the assembly is *not* replaced until the third day of evaluation? (Assume the daily decisions are independent and use the geometric

distribution.) (c) Suppose on the first day of evaluation two of the blades are dull, on the second day six are dull, and on the third day ten are dull. What is the probability that the assembly is not replaced until the third day of evaluation? (Assume the daily decisions are independent; here the probability of replacement changes every day.)

Solution. Let X denote the number of dull blades in the sample of five (selected from 48).

(a) With $K = 10$, $N - K = 38$, $n = 5$,

$$\mathrm{P}(X = 0) = \frac{\binom{10}{0}\binom{38}{5}}{\binom{48}{5}} = \frac{38!\,/\,(33!\,5!)}{48!\,/\,(43!\,5!)} = 0.2931,$$

$$\mathrm{P}(\text{replaced}) = \mathrm{P}(X \geq 1) = 1 - \mathrm{P}(X = 0) = 0.7069.$$

(b) Let Y denote the number of days needed to replace the assembly. Each day, replacement occurs with probability $p = 0.7069$, so Y is geometric:

$$\mathrm{P}(Y = 3) = (1 - p)^{y-1}p = (1 - 0.7069)^2(0.7069) = 0.0607.$$

(c) The probability of *no* dull blade in the sample changes each day. On the first day ($K = 2$):

$$\mathrm{P}(X = 0) = \frac{\binom{2}{0}\binom{46}{5}}{\binom{48}{5}} = \frac{46!\,/\,(41!\,5!)}{48!\,/\,(43!\,5!)} = 0.8005.$$

On the second day ($K = 6$):

$$\mathrm{P}(X = 0) = \frac{\binom{6}{0}\binom{42}{5}}{\binom{48}{5}} = \frac{42!\,/\,(37!\,5!)}{48!\,/\,(43!\,5!)} = 0.4968.$$

On the third day ($K = 10$): $\mathrm{P}(X = 0) = 0.2931$. Therefore the probability that the assembly is not replaced until the third day is

$$\mathrm{P}(\text{not replaced until 3rd day}) = 0.8005 \times 0.4968 \times (1 - 0.2931) = 0.2811.$$

Chapter 3

The Poisson Distribution

The Poisson distribution models the number of events (counts) occurring in a fixed interval of time, length, area, or volume, when the events occur at a constant average rate and independently of one another. Its single parameter is the mean number of events λ in the interval of interest.

3.1 Definition

Definition 3.1: Poisson random variable

The random variable X that equals the number of events in a Poisson process is a *Poisson* random variable with parameter $\lambda > 0$, and its probability mass function is

$$f(x) = \frac{e^{-\lambda}\lambda^x}{x!}, \qquad x = 0, 1, 2, \ldots$$

Key Formula

For a Poisson random variable with parameter λ,

$$\mathbb{E}(X) = \lambda, \qquad \mathrm{Var}(X) = \lambda.$$

The mean and the variance are equal. If the rate is given per unit interval and an interval of size t is considered, the mean over that interval is λt.

3.2 Worked Problems

Example 3.1 (Messages to a bulletin board)**.** The number of messages sent to a computer bulletin board is a Poisson random variable with a mean of 5 messages per hour. (a) What is the probability that 5 messages are received in 1 hour? (b) What is the probability that 10 messages are received in 1.5 hours? (c) What is the probability that fewer than 2 messages are received in one-half hour?

Solution.

(a) Let X denote the number of messages sent in 1 hour, so $\lambda = 5$:

$$\mathrm{P}(X = 5) = \frac{e^{-5}5^5}{5!} = 0.1755.$$

(b) Let Y denote the number of messages sent in 1.5 hours. Then Y is Poisson with $\lambda = 5 \times 1.5 =$

7.5 messages per 1.5 hours:

$$\mathrm{P}(Y = 10) = \frac{e^{-7.5}(7.5)^{10}}{10!} = 0.0858.$$

(c) Let W denote the number of messages sent in one-half hour. Then W is Poisson with $\lambda = 2.5$:

$$\mathrm{P}(W < 2) = \mathrm{P}(W = 0) + \mathrm{P}(W = 1) = \frac{e^{-2.5}(2.5)^0}{0!} + \frac{e^{-2.5}(2.5)^1}{1!} = 0.2878.$$

Example 3.2 (Errors in a textbook)**.** The number of errors in a textbook follows a Poisson distribution with a mean of 0.01 error per page. What is the probability that there are three or fewer errors in 100 pages?

Solution. The number of errors per page is Poisson with $\lambda = 0.01$. Over 100 pages the mean is $\lambda = 0.01 \times 100 = 1$, so let Y be Poisson with $\lambda = 1$:

$$\mathrm{P}(Y \leq 3) = e^{-1} + \frac{e^{-1}(1)^1}{1!} + \frac{e^{-1}(1)^2}{2!} + \frac{e^{-1}(1)^3}{3!} = 0.9810.$$

3.3 A Cumulative Distribution Function Exercise

The following exercise determines the probability mass function of a discrete random variable from its cumulative distribution function.

Example 3.3 (pmf from a given cdf)**.** Determine the probability mass function for the random variable with the following cumulative distribution function:

$$F(x) = \begin{cases} 0 & x < 2, \\ 0.2 & 2 \leq x < 5.7, \\ 0.5 & 5.7 \leq x < 6.5, \\ 0.8 & 6.5 \leq x < 8.5, \\ 1 & 8.5 \leq x. \end{cases}$$

Solution. The probability mass function places a jump at each point of increase of F, equal to the size of the jump:

$$f(x) = \begin{cases} 0.2 & x = 2, \\ 0.3 & x = 5.7, \\ 0.3 & x = 6.5, \\ 0.2 & x = 8.5, \\ 0 & \text{otherwise.} \end{cases}$$

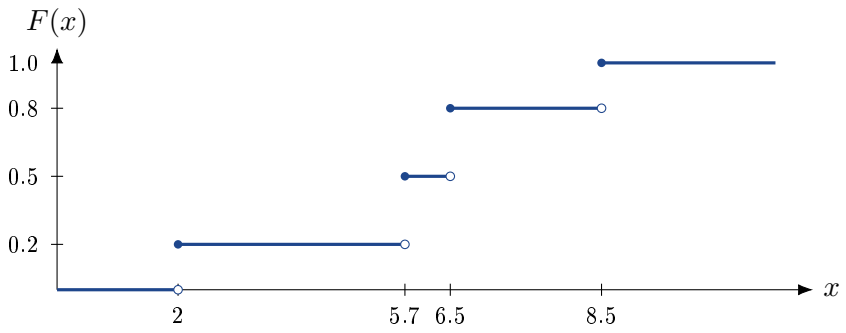


Figure 3.1: Step-function cumulative distribution function $F(x)$; the jump heights give the probability mass function.

Chapter 4

Continuous Random Variables and Probability Density Functions

We now move from discrete to continuous random variables. A continuous random variable can take any value in an interval of the real line, and its probability distribution is described not by a mass function but by a *probability density function*.

4.1 Probability Density Functions

A probability density function $f(x)$ can be used to describe the probability distribution of a continuous random variable X. If an interval is likely to contain a value for X, its probability is large, and it corresponds to large values of $f(x)$. The probability that X is between a and b is determined as the integral of $f(x)$ from a to b.

Definition 4.1: Probability density function

For a continuous random variable X, a *probability density function* is a function $f(x)$ such that

(1) $f(x) \geq 0$;

(2) $\displaystyle\int_{-\infty}^{\infty} f(x)\,dx = 1$;

(3) $\mathrm{P}(a \leq X \leq b) = \displaystyle\int_a^b f(x)\,dx$ = area under $f(x)$ from a to b, for any a and b.

A probability density function is zero for x values that cannot occur, and it is assumed to be zero wherever it is not specifically defined. A histogram is an approximation to a probability density function.

For a continuous random variable X and any value x, $\mathrm{P}(X = x) = 0$: the probability density at any single point is, in this sense, "zero area." Consequently, for any x_1 and x_2,

$$\mathrm{P}(x_1 \leq X \leq x_2) = \mathrm{P}(x_1 < X \leq x_2) = \mathrm{P}(x_1 \leq X < x_2) = \mathrm{P}(x_1 < X < x_2).$$

4.2 Worked Problems on Probability Density Functions

Example 4.1 (An exponential-type density)**.** Suppose that $f(x) = e^{-(x-4)}$ for $4 < x$. Determine the following probabilities: (a) $\mathrm{P}(1 < X)$, (b) $\mathrm{P}(2 \leq X < 5)$, (c) $\mathrm{P}(5 < X)$, (d) $\mathrm{P}(8 < X < 12)$, and (e) determine x such that $\mathrm{P}(X < x) = 0.90$.

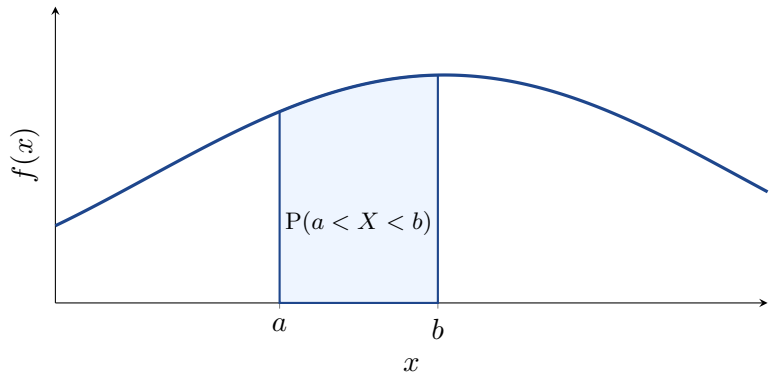


Figure 4.1: The probability that X lies between a and b is the shaded area under the density curve.

Solution.

$$\text{(a)}\quad \mathrm{P}(1 < X) = \int_4^\infty e^{-(x-4)}\,dx = -e^{-(x-4)}\Big|_4^\infty = 1, \quad \text{since } f(x) = 0 \text{ for } x < 4.$$

$$\text{(b)}\quad \mathrm{P}(2 \le X < 5) = \int_4^5 e^{-(x-4)}\,dx = -e^{-(x-4)}\Big|_4^5 = 1 - e^{-1} = 0.6321.$$

$$\text{(c)}\quad \mathrm{P}(5 < X) = 1 - \mathrm{P}(X \le 5) = 1 - 0.6321 = 0.3679.$$

$$\text{(d)}\quad \mathrm{P}(8 < X < 12) = \int_8^{12} e^{-(x-4)}\,dx = -e^{-(x-4)}\Big|_8^{12} = e^{-4} - e^{-8} = 0.0180.$$

$$\text{(e)}\quad \mathrm{P}(X < x) = \int_4^x e^{-(x-4)}\,dx = 1 - e^{-(x-4)} = 0.90 \;\Rightarrow\; x = 4 - \ln(0.10) = 6.303.$$

Example 4.2 (A linear density)**.** Suppose that $f(x) = x/8$ for $3 < x < 5$. Determine the following probabilities: (a) $\mathrm{P}(X < 4)$, (b) $\mathrm{P}(X > 3.5)$, (c) $\mathrm{P}(4 < X < 5)$, (d) $\mathrm{P}(X < 4.5)$, and (e) $\mathrm{P}(X < 3.5 \text{ or } X > 4.5)$.

Solution.

$$\text{(a)}\quad \mathrm{P}(X < 4) = \int_3^4 \frac{x}{8}\,dx = \frac{x^2}{16}\Big|_3^4 = \frac{16-9}{16} = \frac{7}{16} = 0.4375.$$

$$\text{(b)}\quad \mathrm{P}(X > 3.5) = \int_{3.5}^5 \frac{x}{8}\,dx = \frac{x^2}{16}\Big|_{3.5}^5 = \frac{25-(3.5)^2}{16} = 0.7969.$$

$$\text{(c)}\quad \mathrm{P}(4 < X < 5) = \int_4^5 \frac{x}{8}\,dx = \frac{x^2}{16}\Big|_4^5 = \frac{25-16}{16} = \frac{9}{16} = 0.5625.$$

$$\text{(d)}\quad \mathrm{P}(X < 4.5) = \int_3^{4.5} \frac{x}{8}\,dx = \frac{x^2}{16}\Big|_3^{4.5} = \frac{(4.5)^2-9}{16} = \frac{11.25}{16} = 0.7031.$$

$$\begin{aligned}\text{(e)}\quad \mathrm{P}(X < 3.5 \text{ or } X > 4.5) &= \int_3^{3.5} \frac{x}{8}\,dx + \int_{4.5}^5 \frac{x}{8}\,dx \\ &= \frac{x^2}{16}\Big|_3^{3.5} + \frac{x^2}{16}\Big|_{4.5}^5 = 0.20315 + 0.296875 = 0.500.\end{aligned}$$

Example 4.3 (A pure exponential density)**.** Suppose that $f(x) = e^{-x}$ for $0 < x$. Determine: (a) $\mathrm{P}(1 < X)$, (b) $\mathrm{P}(1 < X < 2.5)$, (c) $\mathrm{P}(X = 3)$, (d) $\mathrm{P}(X < 4)$, (e) $\mathrm{P}(3 \le X)$, (f) determine x such that $\mathrm{P}(x < X) = 0.10$, and (g) determine x such that $\mathrm{P}(X \le x) = 0.10$.

Solution.

$$\text{(a)}\quad \mathrm{P}(1 < X) = \int_1^\infty e^{-x}\,dx = -e^{-x}\big|_1^\infty = e^{-1} = 0.3679.$$

$$\text{(b)}\quad \mathrm{P}(1 < X < 2.5) = \int_1^{2.5} e^{-x}\,dx = -e^{-x}\big|_1^{2.5} = e^{-1} - e^{-2.5} = 0.2858.$$

$$\text{(c)}\quad \mathrm{P}(X = 3) = 0.$$

$$\text{(d)}\quad \mathrm{P}(X < 4) = \int_0^4 e^{-x}\,dx = -e^{-x}\big|_0^4 = 1 - e^{-4} = 0.9816.$$

$$\text{(e)}\quad \mathrm{P}(3 \le X) = \int_3^\infty e^{-x}\,dx = e^{-3} = 0.0498.$$

$$\text{(f)}\quad \mathrm{P}(x < X) = \int_x^\infty e^{-x'}\,dx' = e^{-x} = 0.10 \;\Rightarrow\; x = 2.30.$$

$$\text{(g)}\quad \mathrm{P}(X \le x) = \int_0^x e^{-x'}\,dx' = 1 - e^{-x} = 0.10 \;\Rightarrow\; e^{-x} = 0.9,$$
$$\ln(0.9) = -x \;\Rightarrow\; x = 0.105.$$

4.3 Cumulative Distribution Functions

Definition 4.2: Cumulative distribution function (continuous)

The *cumulative distribution function* of a continuous random variable X is

$$F(x) = \mathrm{P}(X \le x) = \int_{-\infty}^x f(u)\,du.$$

Example 4.4 (A piecewise-linear cdf)**.** Suppose the cdf of the random variable X is

$$F(x) = \begin{cases} 0 & x < 0, \\ 0.2x & 0 \le x < 5, \\ 1 & 5 \le x. \end{cases}$$

Determine: (a) $\mathrm{P}(X < 2.8)$, (b) $\mathrm{P}(X > 1.5)$, (c) $\mathrm{P}(X < -2)$, (d) $\mathrm{P}(X > 6)$.

Solution.

(a) $\mathrm{P}(X < 2.8) = 0.2 \times 2.8 = 0.56.$

(b) $\mathrm{P}(X > 1.5) = 1 - \mathrm{P}(X < 1.5) = 1 - 0.2 \times 1.5 = 1 - 0.3 = 0.7.$

(c) $\mathrm{P}(X < -2) = 0.$

(d) $\mathrm{P}(X > 6) = 0.$

Example 4.5 (A cdf with a slope and an offset)**.** The cdf of a random variable X is

$$F(x) = \begin{cases} 0 & x < -2, \\ 0.25x + 0.5 & -2 \le x < 2, \\ 1 & 2 \le x. \end{cases}$$

Determine: (a) $\mathrm{P}(X < 1.8)$, (b) $\mathrm{P}(X > -1.5)$, (c) $\mathrm{P}(X < -2)$, (d) $\mathrm{P}(-1 < X < 1)$.

Solution.

$$\begin{aligned}
\text{(a)}\quad & \mathrm{P}(X < 1.8) = 0.25 \times 1.8 + 0.5 = 0.95.\\
\text{(b)}\quad & \mathrm{P}(X > -1.5) = 1 - \mathrm{P}(X < -1.5) = 1 - (0.25 \times (-1.5) + 0.5) = 0.875.\\
\text{(c)}\quad & \mathrm{P}(X < -2) = 0.\\
\text{(d)}\quad & \mathrm{P}(-1 < X < 1) = \mathrm{P}(X < 1) - \mathrm{P}(X \le -1)\\
& \qquad = (0.25 \times 1 + 0.5) - (0.25 \times (-1) + 0.5)\\
& \qquad = 0.75 - 0.25 = 0.5.
\end{aligned}$$

Example 4.6 (Gap width of a magnetic recording head)**.** The gap width is an important property of a magnetic recording head. In coded units, if the width is a continuous random variable over the range $0 < x < 2$ with $f(x) = 0.5x$, determine the cumulative distribution function of the gap width.

Solution.

$$F(x) = \int_0^x 0.5u\, du = 0.25x^2, \qquad 0 \le x < 2,$$

so that

$$F(x) = \begin{cases} 0 & x < 0,\\ 0.25x^2 & 0 \le x < 2,\\ 1 & 2 \le x. \end{cases}$$

4.4 Mean and Variance of a Continuous Random Variable

Key Formula

For a continuous random variable X with probability density function $f(x)$, the *mean* or expected value is

$$\mu = \mathbb{E}(X) = \int_{-\infty}^{\infty} x\, f(x)\, dx,$$

and the *variance* is

$$\sigma^2 = \mathrm{Var}(X) = \int_{-\infty}^{\infty} (x-\mu)^2 f(x)\, dx = \int_{-\infty}^{\infty} x^2 f(x)\, dx - \mu^2.$$

The standard deviation of X is $\sigma = \sqrt{\sigma^2}$.

Example 4.7 (Uniform density on $(0, 4)$)**.** Suppose $f(x) = 0.25$ for $0 < x < 4$. Determine the mean and variance of X.

Solution.

$$\begin{aligned}
\mathbb{E}(X) &= \int_0^4 x(0.25)\, dx = 0.25\frac{x^2}{2}\Big|_0^4 = 0.25 \times 8 = 2.\\
\mathrm{Var}(X) &= \int_0^4 x^2(0.25)\, dx - \mu^2 = 0.25\frac{x^3}{3}\Big|_0^4 - 2^2 = 0.25 \times \frac{64}{3} - 4\\
&= \frac{16}{3} - 4 = \frac{16-12}{3} = \frac{4}{3}.
\end{aligned}$$

Example 4.8 (Thickness of a conductive coating)**.** The thickness of a conductive coating in micrometres has a density function $600x^{-2}$ for $100\,\mu\text{m} < x < 120\,\mu\text{m}$. (a) Determine the mean and variance of the coating thickness. (b) If the coating costs \$0.50 per micrometre of thickness on each part, what is the average cost of coating per part?

Solution. Using $\int x^{-1}\,dx = \ln x + C$,

$$\begin{aligned}
\mathbb{E}(X) &= \int_{100}^{120} x \cdot \frac{600}{x^2}\,dx = \int_{100}^{120} \frac{600}{x}\,dx = 600\big[\ln(120) - \ln(100)\big] = 109.39.\\
\text{Var}(X) &= \int_{100}^{120} x^2 \cdot \frac{600}{x^2}\,dx - \mu^2 = \int_{100}^{120} 600\,dx - (109.39)^2\\
&= 600\big[120 - 100\big] - (109.39)^2 = 12000 - 11966.1721 = 33.8279.
\end{aligned}$$

(b) Total cost = mean $\times$ cost per μm $= 109.39 \times 0.5 = \$54.70$ per part.

Example 4.9 (Length of computer cables)**.** Suppose the pdf of the length of computer cables is $f(x) = 0.1$ from 1200 to 1210 mm. (a) Determine the mean and standard deviation of the cable length. (b) If the length specifications are $1195 < x < 1205$ mm, what proportion of cables are within specifications?

Solution.

$$\begin{aligned}
\mathbb{E}(X) &= \int_{1200}^{1210} x(0.1)\,dx = 0.05x^2\big|_{1200}^{1210} = 1205.\\
\text{Var}(X) &= \int_{1200}^{1210} (x - 1205)^2(0.1)\,dx = 0.1\frac{(x-1205)^3}{3}\bigg|_{1200}^{1210} = 8.333,\\
\sigma &= \sqrt{8.333} = 2.887.
\end{aligned}$$

(b) Clearly, centring the process at the centre of the specifications results in the greatest proportion of cables within specifications. Since the density is zero below 1200,

$$\text{P}(1195 < X < 1205) = \text{P}(1200 < X < 1205) = \int_{1200}^{1205} 0.1\,dx = 0.1x|_{1200}^{1205} = 0.5.$$

Chapter 5

The Continuous Uniform Distribution

The simplest continuous distribution is one in which the density is constant over a finite interval. It is the continuous analogue of equally likely outcomes.

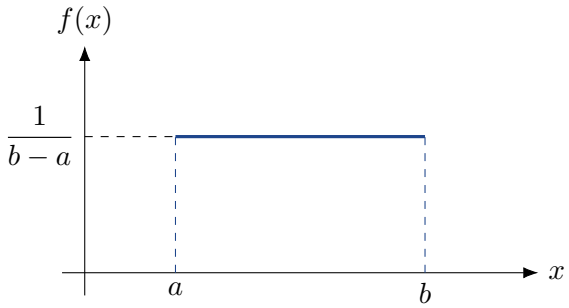
5.1 Definition and Moments

Definition 5.1: Continuous uniform random variable

A continuous random variable X with probability density function

$$f(x) = \frac{1}{b-a}, \qquad a \leq x \leq b,$$

is a *continuous uniform* random variable.



Continuous uniform PDF

Figure 5.1: The continuous uniform probability density function on $[a, b]$.

Key Formula

For a continuous uniform random variable on $[a, b]$,

$$\mu = \mathbb{E}(X) = \frac{a+b}{2}, \qquad \sigma^2 = \mathrm{Var}(X) = \frac{(b-a)^2}{12},$$

and the cumulative distribution function is

$$F(x) = \begin{cases} 0 & x < a, \\ \dfrac{x-a}{b-a} & a \leq x < b, \\ 1 & b \leq x. \end{cases}$$

These follow from

$$\mathbb{E}(X) = \int_a^b \frac{x}{b-a}\,dx = \left.\frac{0.5x^2}{b-a}\right|_a^b = \frac{a+b}{2},$$

$$\mathrm{Var}(X) = \int_a^b \frac{\left(x - \frac{a+b}{2}\right)^2}{b-a}\,dx = \left.\frac{\left(x - \frac{a+b}{2}\right)^3}{3(b-a)}\right|_a^b = \frac{(b-a)^2}{12},$$

and

$$F(x) = \int_a^x \frac{1}{b-a}\,du = \frac{x-a}{b-a}, \qquad a \le x < b.$$

5.2 Worked Problems

Example 5.1 (Net weight of a packaged herbicide)**.** The net weight in pounds of a packaged chemical herbicide is uniform for $49.75 < x < 50.25$ pounds. (a) Determine the mean and variance of the weight of the packages. (b) Determine the cdf of the weight of the packages. (c) Determine $\mathrm{P}(X < 50.1)$.

Solution. Here $a = 49.75$, $b = 50.25$, so $b - a = 0.5$ and $f(x) = 1/0.5 = 2$ for $49.75 < x < 50.25$.

(a)

$$\mathbb{E}(X) = \frac{a+b}{2} = \frac{49.75 + 50.25}{2} = 50, \qquad \mathrm{Var}(X) = \frac{(50.25 - 49.75)^2}{12} = 0.0208,$$

with $\sigma = \sqrt{0.0208} = 0.144$.

(b)

$$F(x) = \begin{cases} 0 & x \le 49.75, \\ \dfrac{x - 49.75}{0.5} = 2x - 99.5 & 49.75 < x < 50.25, \\ 1 & x \ge 50.25. \end{cases}$$

(c) $\mathrm{P}(X < 50.1) = F(50.1) = 2 \times 50.1 - 99.5 = 0.7$.

Example 5.2 (Thickness of an aircraft flange)**.** The thickness of a flange on an aircraft component is uniformly distributed between 0.95 and 1.05 mm. (a) Determine the cdf of the flange thickness. (b) Determine the proportion of flanges that exceed 1.02 mm. (c) What thickness is exceeded by 90% of the flanges? (d) Determine the mean and variance of flange thickness.

Solution. Here $a = 0.95$, $b = 1.05$, so $f(x) = 1/0.1 = 10$ for $0.95 < x < 1.05$.

(a)

$$F(x) = \begin{cases} 0 & x < 0.95, \\ \dfrac{x - 0.95}{1.05 - 0.95} = 10x - 9.5 & 0.95 \le x < 1.05, \\ 1 & 1.05 \le x. \end{cases}$$

(b)

$$\mathrm{P}(X > 1.02) = 1 - \mathrm{P}(X \le 1.02) = 1 - F(1.02) = 1 - (10 \times 1.02 - 9.5) = 1 - 0.7 = 0.3.$$

(c) If $\mathrm{P}(X > x) = 0.9$ then $1 - F(x) = 0.90$ and $F(x) = 0.10$, so

$$10x - 9.5 = 0.10 \;\Rightarrow\; x = 0.96.$$

(d)

$$\mathbb{E}(X) = \frac{1.05 + 0.95}{2} = 1.00, \qquad \mathrm{Var}(X) = \frac{(1.05 - 0.95)^2}{12} = 0.00083.$$

Example 5.3 (Time to fill an electronic form)**.** Suppose the time it takes a data collection operator to fill out an electronic form for a database is uniformly distributed between 1.5 and 2.2 minutes. (a) What is the mean and variance of the time it takes an operator to fill out the form? (b) What is the probability that it will take less than 2 minutes to fill out the form? (c) Determine the cdf of the time it takes to fill out the form.

Solution. Here $a = 1.5$, $b = 2.2$, so $b - a = 0.7$ and $f(x) = 1/0.7$.

(a)

$$\mathbb{E}(X) = \frac{1.5 + 2.2}{2} = 1.85 \text{ min}, \qquad \mathrm{Var}(X) = \frac{(2.2 - 1.5)^2}{12} = 0.0408 \text{ min}^2.$$

(b)

$$\mathrm{P}(X < 2) = \int_{1.5}^{2} \frac{1}{0.7}\, dx = \frac{x}{0.7}\Big|_{1.5}^{2} = \frac{2 - 1.5}{0.7} = \frac{0.5}{0.7} = 0.7143.$$

(c)

$$F(x) = \begin{cases} 0 & x < 1.5, \\ \dfrac{x - 1.5}{0.7} & 1.5 \le x < 2.2, \\ 1 & 2.2 \le x. \end{cases}$$

Example 5.4 (Thickness of photoresist)**.** The thickness of photoresist applied to wafers in semiconductor manufacturing at a particular location on the wafer is uniformly distributed between 0.2050 and 0.2150 micrometres. (a) Determine the cdf of photoresist thickness. (b) Determine the proportion of wafers that exceed 0.2125 μm in photoresist thickness. (c) What thickness is exceeded by 10% of the wafers? (d) Determine the mean and variance of photoresist thickness.

Solution. The pdf is

$$f(x) = \frac{1}{b - a} = \frac{1}{0.2150 - 0.2050} = \frac{1}{0.01} = 100, \qquad 0.2050 < x < 0.2150.$$

(a)

$$F(x) = \begin{cases} 0 & x < 0.2050, \\ \dfrac{x - 0.2050}{0.0100} = 100x - 20.50 & 0.2050 \le x < 0.2150, \\ 1 & 0.2150 \le x. \end{cases}$$

(b)

$$\mathrm{P}(X > 0.2125) = 1 - F(0.2125) = 1 - \big[100(0.2125) - 20.50\big] = 0.25.$$

(c) If $\mathrm{P}(X > x) = 0.10$ then $1 - F(x) = 0.10$ and $F(x) = 0.90$, so

$$100x + 20.50 = 0.90 \;\Rightarrow\; (\text{solving } 100x - 20.50 = 0.90)\;\; x = 0.2140.$$

(d)

$$\mathbb{E}(X) = \frac{0.2050 + 0.2150}{2} = 0.2100 \ \mu\text{m}, \qquad \mathrm{Var}(X) = \frac{(0.2150 - 0.2050)^2}{12} = 8.33 \times 10^{-6} \ \mu\text{m}^2.$$

Chapter 6

The Normal Distribution

The most widely used model for the distribution of a random variable is the *normal distribution*. Whenever a random experiment is replicated, the random variable that equals the average (or total) result over the replicates tends to have a normal distribution as the number of replicates becomes large. De Moivre presented this fundamental result, known as the *central limit theorem*, in 1733. About a hundred years later Gauss independently developed a normal distribution; a normal distribution is also referred to as a *Gaussian distribution*.

6.1 The Normal Probability Density Function

Random variables with different means and variances can be modelled by normal probability density functions with appropriate choices of the centre and width of the curve. The value of $\mathbb{E}(X) = \mu$ determines the centre of the probability density function, and the value of $\text{Var}(X) = \sigma^2$ determines the width.

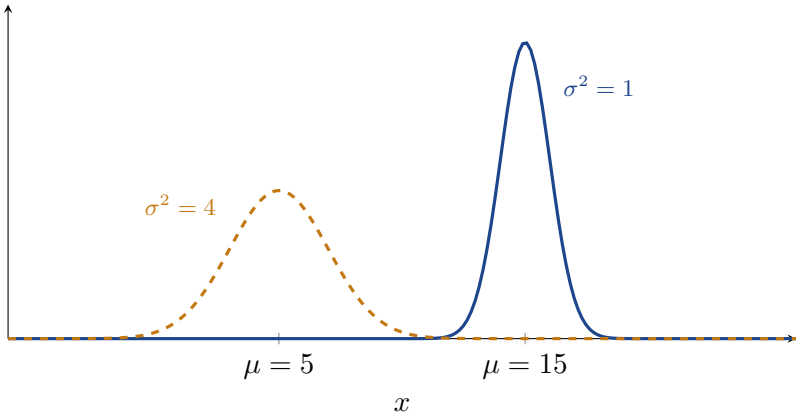


Figure 6.1: Normal densities with different means and variances. The mean locates the centre; the variance controls the spread.

Definition 6.1: Normal random variable

A random variable X with probability density function

$$f(x) = \frac{1}{\sqrt{2\pi}\,\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \qquad -\infty < x < \infty,$$

is a *normal* random variable with parameters μ (where $-\infty < \mu < \infty$) and $\sigma > 0$. Also $\mathbb{E}(X) = \mu$ and $\text{Var}(X) = \sigma^2$, and the notation $N(\mu, \sigma^2)$ is used to denote the distribution.

6.2 The Standard Normal Distribution

Definition 6.2: Standard normal random variable

A normal random variable with $\mu = 0$ and $\sigma^2 = 1$ is called a *standard normal* random variable and is denoted as Z. The cumulative distribution function of a standard normal random variable is denoted

$$\Phi(z) = \mathrm{P}(Z \leq z).$$

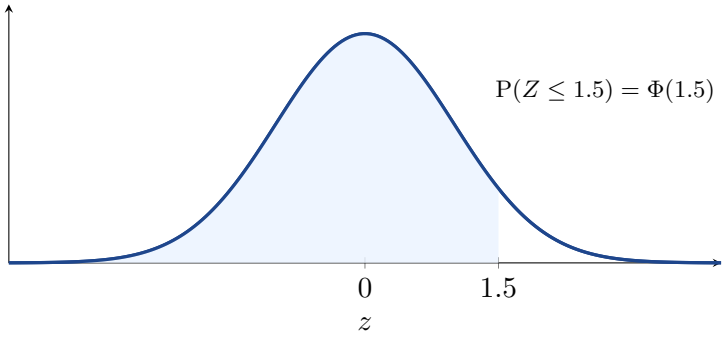


Figure 6.2: The standard normal density; the shaded area to the left of $z = 1.5$ is $\Phi(1.5)$.

A small excerpt of the standard normal table is reproduced below.

Table 6.1: Excerpt of the standard normal cumulative probabilities $\Phi(z)$.

z	0.00	0.01	0.02	0.03
0.0	0.50000	0.50399	0.50398	0.51197
⋮				
1.5	0.93319	0.93448	0.93574	0.93699

Probabilities that are not of the form $\mathrm{P}(Z \leq z)$ are found by using the basic rules of probability and the symmetry of the normal distribution along with the tables.

Example 6.1 (Reading the standard normal table). 1) $\mathrm{P}(Z > 1.26) = 1 - \mathrm{P}(Z \leq 1.26) = 1 - 0.89616 = 0.10384$.

2) $\mathrm{P}(Z < -0.86) = 0.19490$.

3) $\mathrm{P}(Z > -1.37) = \mathrm{P}(Z < 1.37) = 0.91465$.

4) $\mathrm{P}(-1.25 < Z < 0.37) = \mathrm{P}(Z < 0.37) - \mathrm{P}(Z < -1.25) = 0.64431 - 0.10565 = 0.53866$.

5) $\mathrm{P}(Z \leq -4.6)$ cannot be found exactly from the table. The last value in the table is $\mathrm{P}(Z \leq -3.99) = 0.00003$. Since $\mathrm{P}(Z \leq -4.6) < \mathrm{P}(Z \leq -3.99)$, we take $\mathrm{P}(Z \leq -4.6) \approx 0$.

6) Find z such that $\mathrm{P}(Z > z) = 0.05$, i.e. $\mathrm{P}(Z \leq z) = 0.95$. The nearest table value is 0.95053, giving $z = 1.65$.

7) Find z such that $\mathrm{P}(-z < Z < z) = 0.99$. By the symmetry of the normal distribution, the area in each tail must equal 0.005. Thus the value for z corresponds to a probability of 0.995 in the table. The nearest probability is 0.99506, occurring at $z = 2.58$.

6.3 Standardizing a Normal Random Variable

Key Formula

If X is a normal random variable with $\mathbb{E}(X) = \mu$ and $\mathrm{Var}(X) = \sigma^2$, then the random variable

$$Z = \frac{X - \mu}{\sigma}$$

is a normal random variable with $\mathbb{E}(Z) = 0$ and $\mathrm{Var}(Z) = 1$; that is, Z is a standard normal random variable. Consequently, for any x,

$$\mathrm{P}(X \le x) = \mathrm{P}\left(\frac{X-\mu}{\sigma} \le \frac{x-\mu}{\sigma}\right) = \mathrm{P}(Z \le z), \qquad z = \frac{x-\mu}{\sigma}.$$

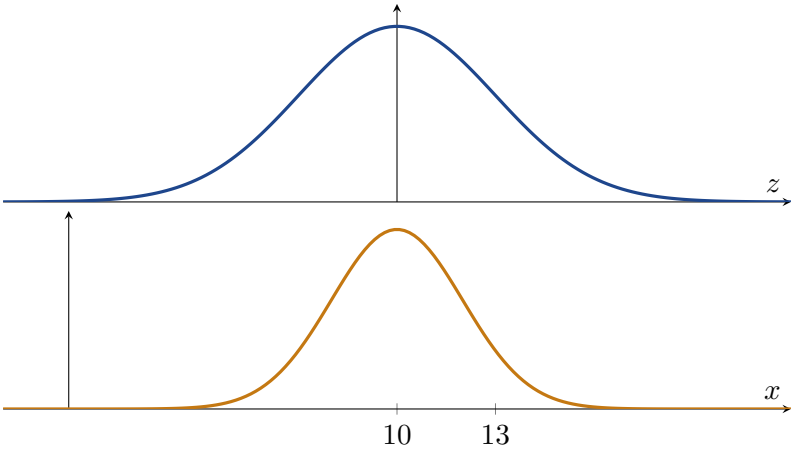


Figure 6.3: Standardizing a normal random variable: the transformation $z = (x - \mu)/\sigma$ maps $X \sim N(\mu, \sigma^2)$ onto the standard normal scale.

6.4 Worked Problems

Example 6.2 (Probabilities for a standard normal variable)**.** Use tables to determine the following probabilities for the standard normal random variable Z: (a) $\mathrm{P}(Z < 1.32)$, (b) $\mathrm{P}(Z < 3.0)$, (c) $\mathrm{P}(Z > 1.45)$, (d) $\mathrm{P}(Z > -2.15)$, (e) $\mathrm{P}(-2.34 < Z < 1.76)$.

Solution.

$$\begin{aligned}
\text{(a)}\quad \mathrm{P}(Z < 1.32) &= 0.906582.\\
\text{(b)}\quad \mathrm{P}(Z < 3.0) &= 0.99865.\\
\text{(c)}\quad \mathrm{P}(Z > 1.45) &= 1 - \mathrm{P}(Z < 1.45) = 1 - 0.92647 = 0.07353.\\
\text{(d)}\quad \mathrm{P}(Z > -2.15) &= \mathrm{P}(Z < 2.15) = 0.98422.\\
\text{(e)}\quad \mathrm{P}(-2.34 < Z < 1.76) &= \mathrm{P}(Z < 1.76) - \mathrm{P}(Z < -2.34)\\
&= 0.9607 - \big(1 - \mathrm{P}(Z < 2.34)\big)\\
&= 0.9607 - (1 - 0.990358)\\
&= 0.9607 - 0.009642 = 0.9511.
\end{aligned}$$

Example 6.3 (A normal variable with mean 10, sd 2)**.** Assume X is normally distributed with

a mean of 10 and a standard deviation of 2. Determine the following: (a) $\mathrm{P}(X < 13)$, (b) $\mathrm{P}(X > 9)$, (c) $\mathrm{P}(6 < X < 14)$, (d) $\mathrm{P}(2 < X < 4)$, (e) $\mathrm{P}(-2 < X < 8)$.

Solution.

$$\begin{aligned}
\text{(a)}\quad \mathrm{P}(X < 13) &= \mathrm{P}\big(Z < \tfrac{13-10}{2}\big) = \mathrm{P}(Z < 1.5) = 0.93319.\\
\text{(b)}\quad \mathrm{P}(X > 9) &= 1 - \mathrm{P}(X < 9) = 1 - \mathrm{P}\big(Z < \tfrac{9-10}{2}\big)\\
&= 1 - \mathrm{P}(Z < -0.5) = 1 - 0.308538 = 0.691462.\\
\text{(c)}\quad \mathrm{P}(6 < X < 14) &= \mathrm{P}\big(\tfrac{6-10}{2} < Z < \tfrac{14-10}{2}\big) = \mathrm{P}(-2 < Z < 2)\\
&= \mathrm{P}(Z < 2) - \mathrm{P}(Z < -2) = 0.977250 - 0.022750 = 0.954500.\\
\text{(d)}\quad \mathrm{P}(2 < X < 4) &= \mathrm{P}\big(\tfrac{2-10}{2} < Z < \tfrac{4-10}{2}\big) = \mathrm{P}(-4 < Z < -3)\\
&= \mathrm{P}(Z < -3) - \mathrm{P}(Z < -4) = 0.001350 - 0.000033 = 0.001327.\\
\text{(e)}\quad \mathrm{P}(-2 < X < 8) &= \mathrm{P}(X < 8) - \mathrm{P}(X < -2) = \mathrm{P}\big(Z < \tfrac{8-10}{2}\big) - \mathrm{P}\big(Z < \tfrac{-2-10}{2}\big)\\
&= \mathrm{P}(Z < -1) - \mathrm{P}(Z < -6) = 0.15866.
\end{aligned}$$

Example 6.4 (Inverse problems for standard normal)**.** Assume Z has a standard normal distribution. Use the table to determine the value for z that solves each of the following: (a) $\mathrm{P}(Z < z) = 0.9$, (b) $\mathrm{P}(Z < z) = 0.5$, (c) $\mathrm{P}(Z > z) = 0.1$, (d) $\mathrm{P}(Z > z) = 0.9$, (e) $\mathrm{P}(-1.24 < Z < z) = 0.8$.

Solution.

$$\begin{aligned}
&\text{(a)}\quad \mathrm{P}(Z < 1.28) = 0.9 \Rightarrow z = 1.28.\\
&\text{(b)}\quad \mathrm{P}(Z < 0) = 0.5 \Rightarrow z = 0.\\
&\text{(c)}\quad \mathrm{P}(Z > z) = 0.1 \Rightarrow \mathrm{P}(Z < z) = 0.9 \Rightarrow z = 1.28.\\
&\text{(d)}\quad \mathrm{P}(Z > z) = 0.9 \Rightarrow \mathrm{P}(Z < z) = 0.10 \Rightarrow z = -1.28.\\
&\text{(e)}\quad \mathrm{P}(-1.24 < Z < z) = 0.8 \Rightarrow \mathrm{P}(Z < z) - \mathrm{P}(Z < -1.24) = 0.8\\
&\qquad \Rightarrow \mathrm{P}(Z < z) - 0.10749 = 0.8 \Rightarrow \mathrm{P}(Z < z) = 0.90749 \Rightarrow z = 1.33.
\end{aligned}$$

Example 6.5 (Inverse problems for a general normal variable)**.** Assume X is normally distributed with a mean of 10 and a standard deviation of 2. Determine the value for x that solves each of the following: (a) $\mathrm{P}(X > x) = 0.5$, (b) $\mathrm{P}(X > x) = 0.95$, (c) $\mathrm{P}(x < X < 10) = 0.2$.

Solution.

$$\begin{aligned}
&\text{(a)}\quad \mathrm{P}(X > x) = 0.5 \Rightarrow \mathrm{P}\big(Z > \tfrac{x-10}{2}\big) = 0.5 \Rightarrow \tfrac{x-10}{2} = 0 \Rightarrow x = 10.\\
&\text{(b)}\quad \mathrm{P}(X > x) = 0.95 \Rightarrow 1 - \mathrm{P}\big(Z < \tfrac{x-10}{2}\big) = 0.95\\
&\qquad \Rightarrow \mathrm{P}\big(Z < \tfrac{x-10}{2}\big) = 0.05 \Rightarrow \tfrac{x-10}{2} = -1.64 \Rightarrow x = 6.72.\\
&\text{(c)}\quad \mathrm{P}(x < X < 10) = 0.2 \Rightarrow \mathrm{P}\big(\tfrac{x-10}{2} < Z < 0\big) = 0.2\\
&\qquad \Rightarrow 0.5 - \mathrm{P}\big(Z < \tfrac{x-10}{2}\big) = 0.2 \Rightarrow \mathrm{P}\big(Z < \tfrac{x-10}{2}\big) = 0.3\\
&\qquad \Rightarrow \tfrac{x-10}{2} = -0.52 \Rightarrow x = 8.96.
\end{aligned}$$

Example 6.6 (Compressive strength of cement)**.** The compressive strength of samples of cement can be modelled by a normal distribution with a mean of 6000 kg/cm^2 and a standard deviation of 100 kg/cm^2. (a) What is the probability that a sample's strength is less than 6250 kg/cm^2?

(b) What is the probability that a sample strength is between 5800 and 5900 kg/cm^2? (c) What strength is exceeded by 95% of the samples?

Solution.

$$\begin{aligned}
\text{(a)}\quad \mathrm{P}(X<6250) &= \mathrm{P}\big(Z<\tfrac{6250-6000}{100}\big) = \mathrm{P}(Z<2.5) = 0.99379.\\
\text{(b)}\quad \mathrm{P}(5800<X<5900) &= \mathrm{P}\big(\tfrac{5800-6000}{100}<Z<\tfrac{5900-6000}{100}\big)\\
&= \mathrm{P}(-2<Z<-1) = \mathrm{P}(Z<-1)-\mathrm{P}(Z<-2) = 0.13591.\\
\text{(c)}\quad \mathrm{P}(X>x)=0.95 &\Rightarrow \mathrm{P}\big(Z>\tfrac{x-6000}{100}\big)=0.95\\
&\Rightarrow \tfrac{x-6000}{100} = -1.65 \Rightarrow x = 5835.
\end{aligned}$$

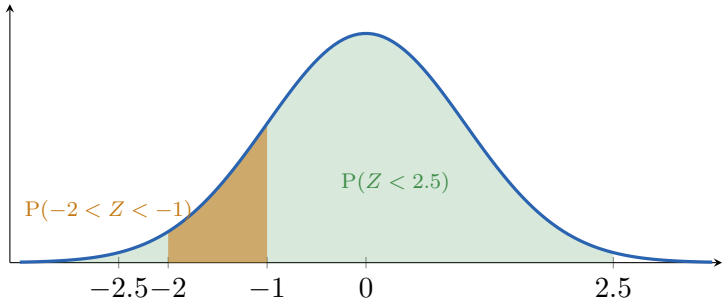


Figure 6.4: Standard-normal areas for the cement-strength problem: part (a) shaded light, part (b) shaded dark.

Example 6.7 (Line width in semiconductor manufacturing)**.** The line width for semiconductor manufacturing is assumed to be normally distributed with a mean of 0.5 μm and a standard deviation of 0.05 μm. (a) What is the probability that a line width is greater than 0.62 μm? (b) What is the probability that a line width is between 0.47 and 0.63 μm? (c) The line width of 90% of samples is below what value?

Solution.

$$\begin{aligned}
\text{(a)}\quad \mathrm{P}(X>0.62) &= \mathrm{P}\big(Z>\tfrac{0.62-0.5}{0.05}\big) = \mathrm{P}(Z>2.4) = 1-\mathrm{P}(Z<2.4) = 0.0082.\\
\text{(b)}\quad \mathrm{P}(0.47<X<0.63) &= \mathrm{P}\big(\tfrac{0.47-0.5}{0.05}<Z<\tfrac{0.63-0.5}{0.05}\big) = \mathrm{P}(-0.6<Z<2.6)\\
&= \mathrm{P}(Z<2.6)-\mathrm{P}(Z<-0.6) = 0.99534-0.27425 = 0.72109.\\
\text{(c)}\quad \mathrm{P}(X<x)=0.90 &\Rightarrow \mathrm{P}\big(Z<\tfrac{x-0.5}{0.05}\big)=0.90\\
&\Rightarrow \tfrac{x-0.5}{0.05} = 1.28 \Rightarrow x = 0.564.
\end{aligned}$$

Example 6.8 (A normal variable with mean 0.4, sd 0.05)**.** Assume X is normally distributed with a mean of 0.4 and a standard deviation of 0.05. (a) Determine $\mathrm{P}(X>0.5)$. (b) Determine $\mathrm{P}(0.4<X<0.5)$. (c) Determine the value x such that $\mathrm{P}(X<x)=0.90$.

Solution.

$$\begin{aligned}
\text{(a)}\quad \mathrm{P}(X>0.5) &= \mathrm{P}\big(Z>\tfrac{0.5-0.4}{0.05}\big) = \mathrm{P}(Z>2) = 0.02275.\\
\text{(b)}\quad \mathrm{P}(0.4<X<0.5) &= \mathrm{P}\big(\tfrac{0.4-0.4}{0.05}<Z<\tfrac{0.5-0.4}{0.05}\big) = \mathrm{P}(0<Z<2) = 0.47725.\\
\text{(c)}\quad \mathrm{P}(X<x)=0.9 &\Rightarrow \mathrm{P}\big(Z<\tfrac{x-0.4}{0.05}\big)=0.90\\
&\Rightarrow \tfrac{x-0.4}{0.05} = 1.28 \Rightarrow x = 0.336 \text{ (i.e. } 0.464).
\end{aligned}$$

Example 6.9 (Speed of a file transfer)**.** The speed of a file transfer from a server on campus to a personal computer at a student's home on a weekday evening is normally distributed with a mean of 60 kilobits/sec and a standard deviation of 4 kilobits/sec. (a) What is the probability that the file will transfer at a speed of 70 kilobits/sec or more? (b) What is the probability that the file will transfer at a speed of less than 58 kb/sec? (c) If the file is 1 MB, what is the average time it will take to transfer the file?

Solution.

$$\text{(a)}\quad \mathrm{P}(X > 70) = \mathrm{P}\big(Z > \tfrac{70-60}{4}\big) = 1 - \mathrm{P}(Z < 2.5) = 1 - 0.99379 = 0.00621.$$

$$\text{(b)}\quad \mathrm{P}(X < 58) = \mathrm{P}\big(Z < \tfrac{58-60}{4}\big) = \mathrm{P}(Z < -0.5) = 0.308538.$$

$$\text{(c)}\quad 1{,}000{,}000 \text{ bytes} \times 8 \text{ bits/byte} = 8{,}000{,}000 \text{ bits},$$

$$\text{average time} = \frac{8{,}000{,}000}{60{,}000} = 133.33 \text{ sec}.$$

Example 6.10 (Life of a semiconductor laser)**.** The life of a semiconductor laser at a constant power is normally distributed with a mean of 7000 hours and a standard deviation of 600 hours. (a) What is the probability that a laser fails before 5000 hours? (b) What is the life in hours that 95% of the lasers exceed? (c) If three lasers are used in a product and they are assumed to fail independently, what is the probability that all three are still operating after 7000 hours?

Solution.

$$\text{(a)}\quad \mathrm{P}(X < 5000) = \mathrm{P}\big(Z < \tfrac{5000-7000}{600}\big) = \mathrm{P}(Z < -3.33) = 0.00043.$$

$$\text{(b)}\quad \mathrm{P}(X > x) = 0.95 \Rightarrow \mathrm{P}\big(Z > \tfrac{x-7000}{600}\big) = 0.95$$

$$\Rightarrow \tfrac{x-7000}{600} = -1.64 \Rightarrow x = 6016.$$

$$\text{(c)}\quad \mathrm{P}(X > 7000) = \mathrm{P}\big(Z > \tfrac{7000-7000}{600}\big) = \mathrm{P}(Z > 0) = 0.5,$$

$$\mathrm{P}(3 \text{ lasers operating after } 7000) = \big(\tfrac{1}{2}\big) \times \tfrac{1}{2} \times \tfrac{1}{2} = \tfrac{1}{8}.$$

Chapter 7

Normal Approximation to the Binomial and Poisson Distributions

Because the binomial and Poisson distributions become difficult to compute when their parameters are large, the normal distribution provides a convenient approximation. This chapter records the standardisation used, the continuity correction, and the rules of thumb that govern when each approximation is adequate.

7.1 Normal Approximation to the Binomial

Key Formula

If X is a binomial random variable with parameters n and p, then

$$Z = \frac{X - np}{\sqrt{np(1-p)}}$$

is approximately a standard normal random variable. To approximate a binomial probability with a normal distribution, a *continuity correction* is applied as follows:

$$\mathrm{P}(X \leq x) = \mathrm{P}(X \leq x + 0.5) \cong \mathrm{P}\left(Z \leq \frac{x + 0.5 - np}{\sqrt{np(1-p)}}\right),$$

$$\mathrm{P}(x \leq X) = \mathrm{P}(x - 0.5 \leq X) \cong \mathrm{P}\left(\frac{x - 0.5 - np}{\sqrt{np(1-p)}} \leq Z\right).$$

The approximation is good for $np > 5$ and $n(1-p) > 5$.

Recall that for a binomial variable X, $\mathbb{E}(X) = np$ and $\mathrm{Var}(X) = np(1-p)$. Consequently, the standardisation equation is nothing more than the formula for standardizing the random variable X. The normal approximation is good when n is large relative to p. A way to remember the continuity correction is to write the probability in terms of $\leq$ or $\geq$ and then add or subtract the 0.5 correction factor to make the probability greater.

7.2 Relationship Among the Hypergeometric, Binomial, and Normal

The binomial distribution is a satisfactory approximation to the hypergeometric distribution when n, the sample size, is small relative to N, the size of the population. A rule of thumb is

that the binomial approximation is effective if $n/N < 0.1$.

For a hypergeometric distribution, p is defined as $p = K/N$, interpreted as the proportion of successes in the population. The normal distribution can in turn provide an effective approximation of hypergeometric probabilities when $n/N < 0.1$, $np > 5$, and $n(1-p) > 5$.

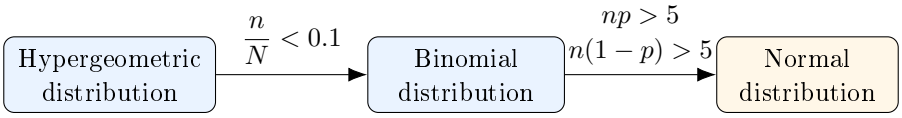


Figure 7.1: Conditions under which one distribution approximates another.

7.3 Normal Approximation to the Poisson

The Poisson distribution was developed as the limit of a binomial distribution as the number of trials increased to infinity. Hence it should not be surprising to find that the normal distribution can also be used to approximate probabilities of a Poisson random variable.

Key Formula

If X is a Poisson random variable with $\mathbb{E}(X) = \lambda$ and $\mathrm{Var}(X) = \lambda$, then

$$Z = \frac{X - \lambda}{\sqrt{\lambda}}$$

is approximately a standard normal random variable. The approximation is good for $\lambda > 5$.

7.4 Worked Problems

Example 7.1 (Normal approximation to a binomial)**.** Suppose that X is a binomial random variable with $n = 200$ and $p = 0.4$. (a) Approximate the probability that X is less than or equal to 70. (b) Approximate the probability that $X > 70$ and $X < 90$. (c) Approximate the probability that $X = 80$.

Solution. First,

$$\mathbb{E}(X) = np = 200(0.4) = 80, \qquad \mathrm{Var}(X) = np(1-p) = 200(0.4)(0.6) = 48, \qquad \sigma = 6.93.$$

(a) With continuity correction,

$$\mathrm{P}(X \le 70) \cong \mathrm{P}\left(Z \le \frac{70.5 - 80}{6.93}\right) = \mathrm{P}(Z \le -1.37) = 0.08534.$$

(Without the continuity correction one would obtain $\mathrm{P}(Z \le \frac{70-80}{\sqrt{48}}) = \mathrm{P}(Z \le -1.44) = 0.0749$.)

(b) With continuity correction, the event $70 < X < 90$ corresponds to $70.5 \le X \le 89.5$:

$$\begin{aligned}\mathrm{P}(70 < X < 90) &\cong \mathrm{P}\left(\frac{70.5 - 80}{6.93} < Z < \frac{89.5 - 80}{6.93}\right) = \mathrm{P}(-1.37 < Z < 1.37) \\ &= \mathrm{P}(Z < 1.37) - \mathrm{P}(Z < -1.37) = 0.91466 - 0.08534 = 0.82932.\end{aligned}$$

(Without the correction, $\mathrm{P}(\frac{70-80}{\sqrt{48}} < Z < \frac{90-80}{\sqrt{48}}) = \mathrm{P}(-1.44 < Z < 1.44)$.)

(c) For a single value, $\mathrm{P}(X = 80) \cong \mathrm{P}(79.5 \le X \le 80.5)$:

$$\mathrm{P}(X = 80) = \mathrm{P}(-0.072 < Z < 0.072) = 0.0558.$$

Example 7.2 (Asbestos particles — normal approximation to Poisson)**.** Assume that the number of asbestos particles in a square metre of dust on a surface follows a Poisson distribution with a mean of 1000. If a square metre of dust is analysed, what is the probability that fewer than 950 particles are found?

Solution. Exactly,

$$\mathrm{P}(X \le 950) = \sum_{x=0}^{950} \frac{e^{-1000}1000^x}{x!},$$

which presents computational difficulty. Using the normal approximation with $\lambda = 1000$ (so $\sigma = \sqrt{1000}$),

$$\mathrm{P}(X \le x) = \mathrm{P}\left(Z \le \frac{950 - 1000}{\sqrt{1000}}\right) = \mathrm{P}(Z \le -1.58) = 0.057.$$

Related question. What is the probability that 10 cm^2 of dust contains more than 10,000 particles? Here $\lambda = 10 \times 1000 = 10{,}000$, so $\mathbb{E}(X) = \lambda = 10^4$ and $\mathrm{Var}(X) = \lambda = 10^4$, giving

$$\mathrm{P}(X > 10{,}000) \cong \mathrm{P}\left(Z > \frac{10{,}000 - 10{,}000}{\sqrt{10{,}000}}\right) = \mathrm{P}(Z > 0) = 0.5.$$

Example 7.3 (Parts failing a qualification test)**.** Fifteen percent of parts fail on being subjected to a qualification test. The parts fail independently of other parts. Three hundred parts are tested every day. Compute the following. (a) The probability that between 35 and 45 parts fail in a day (excluding the endpoints). (b) The probability that there are at most 25 part failures in a day. (c) Of 10 consecutive days when parts are tested, what is the probability that there were exactly 3 days with 25 or fewer part failures?

Solution. With $p = 0.15$ and $n = 300$,

$$\mathbb{E}(X) = np = 300(0.15) = 45, \quad \mathrm{Var}(X) = np(1-p) = 300(0.15)(0.85) = 38.25, \quad \sigma = 6.185.$$

(a) Using the continuity correction for the open interval $35 < X < 45 \Rightarrow 35.5 \le X \le 44.5$:

$$\begin{aligned}\mathrm{P}(35.5 \le X \le 44.5) &= \mathrm{P}\left(\frac{35.5 - 45}{6.185} \le Z \le \frac{44.5 - 45}{6.185}\right) \\ &= \mathrm{P}(-1.536 \le Z \le -0.08) = 0.468 - 0.061 = 0.407.\end{aligned}$$

(b)

$$\mathrm{P}(X \le 25) \cong \mathrm{P}\left(Z \le \frac{24.5 - 45}{6.185}\right) = \mathrm{P}(Z \le -3.31) = 0.000434.$$

(c) The number D of days (out of 10) with 25 or fewer failures is Bin(10, 0.0004). Hence

$$\mathrm{P}(D = 3) = \binom{10}{3}(0.0004)^3(0.9996)^7 \approx 0,$$

an extremely small probability (essentially negligible).

Example 7.4 (Minor print-quality errors)**.** A high-volume printer produces minor print-quality errors on a test pattern of 1000 pages of text according to a Poisson distribution with a mean of 0.4 per page. (a) Why are the numbers of errors on each page independent random variables? (b) What is the mean number of pages with one or more errors? (c) Approximate the probability that more than 350 pages contain errors (one or more).

Solution. Let X represent the number of minor errors on one page of the test pattern. X is a Poisson random variable with mean 0.4 per page.

(a) The numbers of errors per page are random variables. The assumption that the occurrence of an event in a subinterval of a Poisson process is independent of events in other subintervals implies that the numbers of events in disjoint intervals are independent. The pages are disjoint intervals, and consequently the error counts per page are independent.

(b) For one page,

$$\mathrm{P}(X=0)=\frac{e^{-0.4}0.4^0}{0!}=0.670, \qquad \mathrm{P}(X\geq 1)=1-\mathrm{P}(X=0)=1-0.670=0.330.$$

The mean number of pages (out of 1000) with one or more errors is

$$1000\times 0.330=330 \text{ pages.}$$

(c) Let Y be the number of pages with errors; Y is binomial with $n=1000$ and $p=0.330$, so $\mathbb{E}(Y)=330$ and $\mathrm{Var}(Y)=1000(0.33)(0.67)$. Then

$$\begin{aligned}\mathrm{P}(Y>350) &\cong \mathrm{P}\left(Z\geq \frac{350-330}{\sqrt{1000(0.33)(0.67)}}\right)=\mathrm{P}(Z\geq 1.35)\\ &=1-\mathrm{P}(Z\leq 1.35)=1-0.9115=0.0885.\end{aligned}$$

Example 7.5 (People who become ill from a bacterium)**.** The percentage of people exposed to a bacterium who become ill is 20%. Assume that people are independent and that 1000 people are exposed to the bacterium. Approximate each of the following. (a) The probability that more than 225 become ill. (b) The probability that between 175 and 225 become ill. (c) The value such that the probability that the number of people who become ill exceeds the value is 0.01.

Solution. With $n=1000$ and $p=0.2$,

$$\mathbb{E}(X)=1000(0.2)=200, \qquad \mathrm{Var}(X)=1000(0.2)(0.8)=160.$$

(a) With continuity correction,

$$\mathrm{P}(X>225)=1-\mathrm{P}\left(Z\leq \frac{225.5-200}{\sqrt{160}}\right)=1-\mathrm{P}(Z\leq 2.015)=1-0.9777=0.0223.$$

(b)

$$\begin{aligned}\mathrm{P}(175\leq X\leq 225) &\cong \mathrm{P}\left(\frac{175.5-200}{\sqrt{160}}\leq Z\leq \frac{224.5-200}{\sqrt{160}}\right)=\mathrm{P}(-1.93\leq Z\leq 1.93)\\ &=\mathrm{P}(Z\leq 1.93)-\mathrm{P}(Z\leq -1.93)=0.973197-0.026803=0.946394.\end{aligned}$$

(c) If $\mathrm{P}(X>x)=0.01$ then

$$\mathrm{P}\left(Z>\frac{(x+0.5)-200}{\sqrt{160}}\right)=0.01 \;\Rightarrow\; \frac{(x+0.5)-200}{\sqrt{160}}=2.33,$$

so $x+0.5=200+29.47$, giving $x=229.5+0.5=230$.

Chapter 8

The Exponential Distribution

The Poisson distribution counts events in a fixed interval. A closely related question concerns the *distance* (in time or space) between successive events. The random variable that measures this distance is the exponential random variable, and it inherits a lack-of-memory property analogous to that of the geometric distribution.

8.1 Definition and Moments

Definition 8.1: Exponential random variable

The random variable X that equals the distance between successive counts of a Poisson process with mean $\lambda > 0$ is an *exponential* random variable with parameter λ. The probability density function of X is

$$f(x) = \lambda e^{-\lambda x}, \qquad 0 \leq x < \infty.$$

The exponential distribution obtains its name from the exponential function in its probability density function.

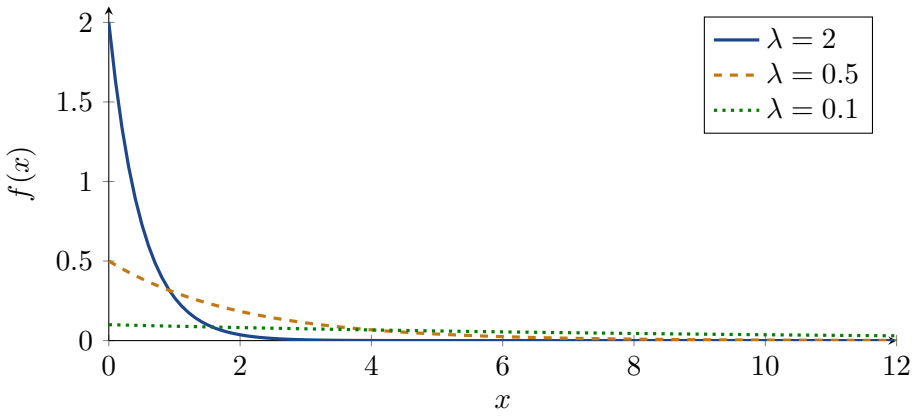


Figure 8.1: Exponential probability density functions for selected values of λ.

Key Formula

If the random variable X has an exponential distribution with parameter λ, then

$$\mu = \mathbb{E}(X) = \frac{1}{\lambda}, \qquad \sigma^2 = \mathrm{Var}(X) = \frac{1}{\lambda^2},$$

and the cumulative distribution function is

$$F(x) = 1 - e^{-\lambda x}, \qquad x \geq 0.$$

It is important to use consistent units in calculations of means, variances, and probabilities involving exponential random variables.

8.2 Worked Problems

Example 8.1 (Exponential with $\lambda = 2$)**.** Suppose X has an exponential distribution with $\lambda = 2$. Determine: (a) $\mathrm{P}(X \le 0)$, (b) $\mathrm{P}(X \ge 2)$, (c) $\mathrm{P}(X \le 1)$, (d) $\mathrm{P}(1 < X < 2)$, (e) find the value of x such that $\mathrm{P}(X < x) = 0.05$.

Solution.

$$\text{(a)}\quad \mathrm{P}(X \le 0) = \int_0^0 \lambda e^{-\lambda x}\,dx = 0.$$

$$\text{(b)}\quad \mathrm{P}(X \ge 2) = \int_2^\infty 2e^{-2x}\,dx = -e^{-2x}\Big|_2^\infty = e^{-4} = 0.0183.$$

$$\text{(c)}\quad \mathrm{P}(X \le 1) = \int_0^1 2e^{-2x}\,dx = -e^{-2x}\Big|_0^1 = 1 - e^{-2} = 0.8647.$$

$$\text{(d)}\quad \mathrm{P}(1 < X < 2) = \int_1^2 2e^{-2x}\,dx = -e^{-2x}\Big|_1^2 = e^{-2} - e^{-4} = 0.1170.$$

$$\text{(e)}\quad \mathrm{P}(X \le x) = \int_0^x 2e^{-2t}\,dt = 1 - e^{-2x} = 0.05 \;\Rightarrow\; x = 0.0256.$$

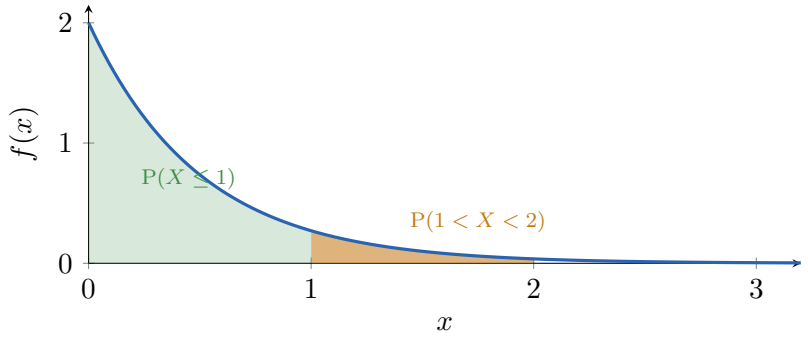


Figure 8.2: Exponential density $f(x) = 2e^{-2x}$ with the areas for parts (c) and (d) shaded.

Example 8.2 (Calls to a plumbing supply business)**.** The time between calls to a plumbing supply business is exponentially distributed with a mean time between calls of 15 minutes. (i) What is the probability that there are no calls within a 30-minute interval? (ii) What is the probability that at least one call arrives within a 10-minute interval? (iii) What is the probability that the first call arrives within 5 and 10 minutes after opening? (iv) Determine the length of an interval of time such that the probability of at least one call in the interval is 0.90.

Solution. The time between calls is exponential with mean 15, so $\lambda = 1/15$.

(i) Either directly,

$$\mathrm{P}(\text{no calls in 30 min}) = \mathrm{P}(T \ge 30) = \int_{30}^\infty \tfrac{1}{15} e^{-x/15}\,dx = -e^{-x/15}\Big|_{30}^\infty = e^{-2} = 0.1353,$$

or via a Poisson variable: if $X \sim \text{Poisson}(30 \cdot \frac{1}{15}) = \text{Poisson}(2)$, then $\mathrm{P}(X = 0) = e^{-2}2^0/0! = 0.1353$.

(ii) Let Y be the number of calls in a 10-minute period, so $Y \sim \text{Poisson}(10 \cdot \frac{1}{15})$:

$$\mathrm{P}(Y \geq 1) = 1 - \mathrm{P}(Y = 0) = 1 - \frac{e^{-2/3}(2/3)^0}{0!} = 1 - e^{-2/3} = 1 - 0.5134 = 0.4866.$$

Equivalently, $\mathrm{P}(X > 10) = e^{-2/3} = 0.5134$ so $\mathrm{P}(X \leq 10) = 0.4866$.

(iii)

$$\mathrm{P}(5 \leq T \leq 10) = \mathrm{P}(T \geq 5) - \mathrm{P}(T \geq 10) = e^{-5/15} - e^{-10/15} = e^{-1/3} - e^{-2/3} = 0.2031.$$

(iv) Let the interval be t. Then W, the number of calls in period t, is Poisson($\frac{1}{15}t$), and

$$\mathrm{P}(W \geq 1) = 0.9 \;\Rightarrow\; 1 - \mathrm{P}(W = 0) = 0.9 \;\Rightarrow\; \mathrm{P}(W = 0) = 0.1.$$

Thus $e^{-t/15} = 0.1$, giving

$$t = -15 \ln(0.1) = 34.5 \text{ minutes.}$$

(The same result follows from $\mathrm{P}(X < x) = 0.9$, i.e. $1 - e^{-x/15} = 0.9$, $x = [-\ln(0.1)] \times 15 = 34.54$ min.)

Example 8.3 (Inter-arrival times in a hotel)**.** Inter-arrival times in minutes between successive customers in a hotel are exponentially distributed with a mean of 2 minutes. (a) Compute the probability that the inter-arrival time between two customers is at most 2 minutes. (b) What is the probability that at least 2 customers arrive in one minute? (c) What is the mean time until every third customer arrives? (d) What is the probability of no customers arriving in a span of 3 minutes?

Solution. Here the mean is 2, so $1/\lambda = 2$ and $\lambda = 0.5$, i.e. $T \sim \text{Exponential}(0.5)$.

(a)

$$\mathrm{P}(T \leq 2) = 1 - e^{-\lambda x} = 1 - e^{-2 \times 0.5} = 1 - e^{-1} = 0.6321.$$

(b) Let X be the number of customers per minute, so $X \sim \text{Poisson}(0.5)$:

$$\begin{aligned}\mathrm{P}(X \geq 2) &= 1 - \big[\mathrm{P}(X = 0) + \mathrm{P}(X = 1)\big] = 1 - \big[e^{-0.5} + e^{-0.5}(0.5)\big] \\ &= 1 - (0.6065 + 0.3033) = 1 - 0.9098 = 0.0901.\end{aligned}$$

(c) The time to the third customer, T', is Erlang$(3, 0.5)$, so

$$\mathbb{E}(T') = \frac{3}{0.5} = 6 \text{ minutes.}$$

(d) Let Y be the number of customers in 3 minutes, so $Y \sim \text{Poisson}(3 \times 0.5) = \text{Poisson}(1.5)$:

$$\mathrm{P}(Y = 0) = \frac{e^{-1.5}(1.5)^0}{0!} = 0.2231.$$

Example 8.4 (Geiger-counter counts)**.** Suppose the counts recorded by a Geiger counter follow a Poisson process with an average of 2 counts per minute. (a) What is the probability that there are no counts in a 30-second interval? (b) What is the probability that the first count occurs in less than 10 seconds? (c) What is the probability that the first count occurs between 1 and 2 minutes after start-up?

Solution. Let X denote the time until the first count. Then X is an exponential random variable with $\lambda = 2$ counts/min.

(a)

$$\mathrm{P}(X > 0.5) = \int_{0.5}^{\infty} 2e^{-2x}\,dx = -e^{-2x}\big|_{0.5}^{\infty} = e^{-1} = 0.3679.$$

(b) Ten seconds = 10/60 min:

$$\mathrm{P}\big(X < \tfrac{10}{60}\big) = \int_{0}^{1/6} 2e^{-2x}\,dx = -e^{-2x}\big|_{0}^{1/6} = 1 - e^{-1/3} = 0.2835.$$

(c)

$$\mathrm{P}(1 < X < 2) = -e^{-2x}\big|_{1}^{2} = e^{-2} - e^{-4} = 0.1170.$$

Example 8.5 (Arrivals of taxis)**.** The time between arrivals of taxis at a busy intersection is exponentially distributed with a mean of 10 minutes. (a) What is the probability that you wait longer than one hour for a taxi? (b) Suppose you have already been waiting for one hour for a taxi; what is the probability that one arrives within the next 10 minutes?

Solution. Let X denote the time until the arrival of a taxi. Then X is exponential with $\lambda = 1/\mathbb{E}(X) = 0.1$ arrivals/min.

(a)

$$\mathrm{P}(X > 60) = \int_{60}^{\infty} 0.1e^{-0.1x}\,dx = -e^{-0.1x}\big|_{60}^{\infty} = e^{-6} = 0.0025.$$

(b) By the lack-of-memory property of the exponential distribution,

$$\mathrm{P}(X < 10) = \int_{0}^{10} 0.1e^{-0.1x}\,dx = -e^{-0.1x}\big|_{0}^{10} = -e^{-1} + 1 = 0.6321.$$

Example 8.6 (Major cracks in a highway)**.** The distance between major cracks in a highway follows an exponential distribution with a mean of 5 miles. (a) What is the probability that there are no major cracks in a 10-mile stretch? (b) What is the probability that there are two major cracks in a 10-mile stretch? (c) What is the standard deviation of the distance between major cracks?

Solution. Let X denote the distance between major cracks. Then X is exponential with

$$\lambda = \frac{1}{\mathbb{E}(X)} = \frac{1}{5} = 0.2 \text{ cracks/mile.}$$

(a)

$$\mathrm{P}(X > 10) = \int_{10}^{\infty} 0.2e^{-0.2x}\,dx = -e^{-0.2x}\big|_{10}^{\infty} = e^{-2} = 0.1353.$$

(b) Let Y denote the number of cracks in 10 miles. Since the distance between cracks is exponential, Y is a Poisson random variable with $\lambda = 10 \times 0.2 = 2$ cracks per 10 miles:

$$\mathrm{P}(Y = 2) = \frac{e^{-2}2^2}{2!} = 0.2707.$$

(c)

$$\sigma_X = \frac{1}{\lambda} = 5 \text{ miles.}$$

Chapter 9

The Erlang and Gamma Distributions

An exponential random variable describes the length until the *first* count is obtained in a Poisson process. A generalisation of the exponential distribution is the length until r counts occur in a Poisson process. The random variable that equals the interval length until r counts occur in a Poisson process is an Erlang random variable.

9.1 The Erlang Distribution

Definition 9.1: Erlang random variable

The random variable X that equals the interval length until r counts occur in a Poisson process with mean λ has an *Erlang* random variable with parameters λ and r. The probability density function of X is

$$f(x) = \frac{\lambda^r x^{r-1} e^{-\lambda x}}{(r-1)!}, \qquad x > 0 \text{ and } r = 1, 2, \ldots$$

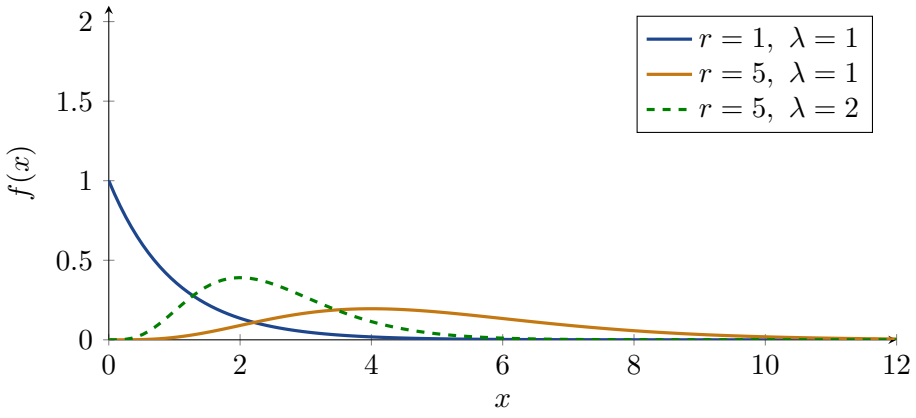


Figure 9.1: Erlang probability density functions for selected r and λ.

Key Formula

If X is an Erlang random variable with parameters λ and r,

$$\mu = \mathbb{E}(X) = \frac{r}{\lambda}, \qquad \sigma^2 = \mathrm{Var}(X) = \frac{r}{\lambda^2}.$$

9.2 The Gamma Distribution

The Erlang distribution is a special case of the gamma distribution. If the parameter r of an Erlang random variable is not an integer but $r > 0$, the random variable has a gamma distribution.

Definition 9.2: The gamma function

The *gamma function* is

$$\Gamma(r) = \int_0^\infty x^{r-1} e^{-x}\, dx, \qquad r > 0.$$

The gamma function has the properties

$$\Gamma(r) = (r-1)\,\Gamma(r-1), \qquad \Gamma(r) = (r-1)! \text{ if } r \text{ is a positive integer (as in Erlang)},$$

$$\Gamma(1) = 0! = 1, \qquad \Gamma\left(\tfrac{1}{2}\right) = \pi^{1/2}.$$

Definition 9.3: Gamma random variable

The random variable X with probability density function

$$f(x) = \frac{\lambda^r x^{r-1} e^{-\lambda x}}{\Gamma(r)}, \qquad x > 0,$$

is a *gamma* random variable with parameters $\lambda > 0$ and $r > 0$. If r is an integer, X has an Erlang distribution.

Key Formula

If X is a gamma random variable with parameters λ and r,

$$\mu = \mathbb{E}(X) = \frac{r}{\lambda}, \qquad \sigma^2 = \mathrm{Var}(X) = \frac{r}{\lambda^2}.$$

The chi-squared distribution is a special case of the gamma distribution, in which $\lambda = \frac{1}{2}$ and r equals one of the values $\frac{1}{2}, 1, \frac{3}{2}, 2, \ldots$ This distribution is used extensively in interval estimation and tests of hypotheses.

9.3 Worked Problems

Example 9.1 (Calls to a telephone system)**.** Calls to a telephone system follow a Poisson distribution with a mean of 5 calls per minute. (a) What is the name applied to the distribution and the parameter values of the time until the tenth call? (b) What is the mean time until the tenth call? (c) What is the mean time between the ninth and tenth calls? (d) What is the probability that exactly four calls occur within one minute? (e) If 10 separate one-minute intervals are chosen, what is the probability that all intervals contain more than two calls?

Solution.

(a) The time until the tenth call is an Erlang random variable with $\lambda = 5$ calls/min and $r = 10$.

(b)

$$\mathbb{E}(X) = \frac{r}{\lambda} = \frac{10}{5} = 2 \text{ min}, \qquad \mathrm{Var}(X) = \frac{r}{\lambda^2} = \frac{10}{25} = 0.4 \text{ min}^2.$$

(c) Because a Poisson process is memoryless, the mean time between the ninth and tenth calls is the mean of an exponential variable,

$$\frac{r}{\lambda} = \frac{1}{5} = 0.2 \text{ min, i.e. 12 seconds.}$$

(d) Let Y denote the number of calls in 1 minute, so $Y \sim \text{Poisson}(5)$:

$$\mathrm{P}(Y = 4) = \frac{e^{-5}5^4}{4!} = 0.1755.$$

(e) Let W denote the number of one-minute intervals (out of 10) that contain more than two calls. For one interval,

$$\mathrm{P}(Y > 2) = 1 - \mathrm{P}(Y \le 2) = 1 - \left[e^{-5} + \frac{e^{-5}5}{1!} + \frac{e^{-5}5^2}{2!}\right] = 0.8754.$$

Because the intervals are independent, W is binomial with $n = 10$ and $p = 0.8754$, so

$$\mathrm{P}(W = 10) = \binom{10}{10}(0.8754)^{10}(1 - 0.8754)^0 = 0.2643.$$

Example 9.2 (Failures of a laser in a cytogenics machine)**.** The time between failures of a laser in a cytogenics machine is exponentially distributed with a mean of 25,000 hours. (a) What is the expected time until the second failure? (b) What is the probability that the time until the third failure exceeds 50,000 hours?

Solution. The time between failures is exponential with mean 25,000, so $1/\lambda = 25{,}000$ and $\lambda = 0.00004$.

(a) The time until the second failure is Erlang with $r = 2$:

$$\mathbb{E}(X) = \frac{r}{\lambda} = \frac{2}{0.00004} = 50{,}000 \text{ hours.}$$

(b) Let N denote the number of failures in 50,000 hours. Then N is Poisson with

$$\mathbb{E}(N) = \frac{50{,}000}{25{,}000} = 2.$$

The time until the third failure exceeds 50,000 hours precisely when fewer than three failures occur in 50,000 hours:

$$\mathrm{P}(N \le 2) = \sum_{k=0}^{2} \frac{e^{-2}2^k}{k!} = 0.6767.$$

Example 9.3 (Time between process problems)**.** The time between process problems in a manufacturing line is exponentially distributed with a mean of 30 days. (a) What is the expected time until the fourth problem? (b) What is the probability that the time until the fourth problem exceeds 120 days?

Solution. Let X denote the time in days until the fourth problem. Then X has an Erlang distribution with $r = 4$ and $\lambda = \frac{1}{30}$ problems/day.

(a)

$$\mathbb{E}(X) = \frac{r}{\lambda} = \frac{4}{1/30} = 120 \text{ days.}$$

(b) Let Y denote the number of problems in 120 days, so $Y \sim \text{Poisson}(120 \cdot \frac{1}{30}) = \text{Poisson}(4)$. The time until the fourth problem exceeds 120 days precisely when fewer than four problems occur:

$$\text{P}(Y \leq 3) = e^{-4} + \frac{e^{-4}4^1}{1!} + \frac{e^{-4}4^2}{2!} + \frac{e^{-4}4^3}{3!} = 0.4335.$$

Example 9.4 (Evaluating the gamma function)**.** Use the gamma function to evaluate the following.

(a) $\Gamma(6) = 5! = 120.$

(b) $\Gamma\left(\frac{5}{2}\right) = \frac{3}{2}\,\Gamma\left(\frac{3}{2}\right) = \frac{3}{2} \cdot \frac{1}{2}\,\Gamma\left(\frac{1}{2}\right) = \frac{3}{4}\,\pi^{1/2} = 1.32934.$

(c) $\Gamma\left(\frac{9}{2}\right) = \frac{7}{2}\,\Gamma\left(\frac{7}{2}\right) = \frac{7}{2} \cdot \frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2}\,\Gamma\left(\frac{1}{2}\right) = \frac{105}{16}\,\pi^{1/2} = 11.6317.$

Example 9.5 (Customers at an ATM)**.** The time between arrivals (TBA) of customers at an automatic teller machine is an exponential random variable with a mean of 5 minutes. (a) What is the probability that more than three customers arrive in 10 minutes? (b) What is the probability that the time until the fifth customer arrives is less than 15 minutes?

Solution. The mean time between arrivals is 5 minutes, so the arrival rate is $\lambda = 0.2$ arrivals/min.

(a) Let X denote the number of customers that arrive in 10 minutes. Then X is Poisson with $\lambda = 0.2 \times 10 = 2$ arrivals per 10 minutes:

$$\begin{aligned}\text{P}(X > 3) &= 1 - \text{P}(X \leq 3) \\ &= 1 - \left[\frac{e^{-2}2^0}{0!} + \frac{e^{-2}2^1}{1!} + \frac{e^{-2}2^2}{2!} + \frac{e^{-2}2^3}{3!}\right] = 0.1429.\end{aligned}$$

(b) Let Y denote the number of customers who arrive in 15 minutes. Then Y is Poisson with $\lambda = 0.2 \times 15 = 3$. The time until the fifth customer is less than 15 minutes precisely when at least five customers arrive in 15 minutes:

$$\begin{aligned}\text{P}(\text{time} < 15) = \text{P}(Y \geq 5) &= 1 - \text{P}(Y \leq 4) \\ &= 1 - \left[e^{-3}\left(1 + 3 + \tfrac{3^2}{2!} + \tfrac{3^3}{3!} + \tfrac{3^4}{4!}\right)\right] = 0.1848.\end{aligned}$$

Chapter 10

The Weibull Distribution

The Weibull distribution is used to model the time until failure of many physical systems. By varying its shape parameter it can represent failure rates that increase, decrease, or remain constant over time, which makes it especially useful in reliability engineering.

10.1 Definition and Moments

Definition 10.1: Weibull random variable

The random variable X with probability density function

$$f(x) = \frac{\beta}{\delta}\left(\frac{x}{\delta}\right)^{\beta-1} \exp\left[-\left(\frac{x}{\delta}\right)^{\beta}\right], \qquad x > 0,$$

is a *Weibull* random variable with scale parameter $\delta > 0$ and shape parameter $\beta > 0$.

If $\beta = 1$, the Weibull distribution is identical to the exponential distribution.

Key Formula

The cumulative distribution function of a Weibull random variable with parameters δ and β is

$$F(x) = 1 - \exp\left[-\left(\frac{x}{\delta}\right)^{\beta}\right],$$

and the mean and variance are

$$\mu = \mathbb{E}(X) = \delta\,\Gamma\left(1 + \frac{1}{\beta}\right),$$

$$\sigma^2 = \mathrm{Var}(X) = \delta^2\,\Gamma\left(1 + \frac{2}{\beta}\right) - \delta^2\left[\Gamma\left(1 + \frac{1}{\beta}\right)\right]^2.$$

10.2 Worked Problem

Example 10.1 (Time to failure of a bearing)**.** The time to failure (in hours) of a bearing in a mechanical shaft is modelled as a Weibull random variable with $\beta = \frac{1}{2}$ and $\delta = 5000$ hours. Determine the mean time until failure, and determine the probability that a bearing lasts at least 6000 hours.

Solution.

$$\mathbb{E}(X) = 5000\,\Gamma\left(1 + \frac{1}{1/2}\right) = 5000\,\Gamma(3) = 5000 \times 2! = 10{,}000 \text{ hours.}$$

$$\mathrm{P}(X > 6000) = 1 - F(6000) = \exp\left[-\left(\frac{6000}{5000}\right)^{1/2}\right] = e^{-1.095} = 0.334.$$

Appendix A

Notes on Reconstruction and Interpretation

In keeping with the goal of reproducing the original handwritten notes as faithfully as possible, the transcription is one-to-one wherever the source was legible. The list below records honestly every place where the handwriting was unclear, an intermediate step was missing, or a minor slip in the original had to be interpreted. In all cases the *final numerical answers written in the original notes were treated as authoritative*, and any reconstructed intermediate working was chosen so as to reach those answers.

1. **Geometric problem, "more than five calls" (Ch. 1).** The original worked $\mathrm{P}(X > 5)$ and separately noted the negative-binomial / "no connection in 5 trials" shortcut. The shortcut was written as $\binom{5}{0}0.02^0 0.98^5$; its stated value 0.9039 matches the tail sum, so both routes are shown.

2. **Uniform density variance $f(x) = 0.25$ on $(0, 4)$ (Ch. 4).** The scanned intermediate line for $\mathrm{Var}(X)$ was partly illegible (a stray "$\frac{16-12}{3}$" fragment). The derivation was reconstructed using $\int_0^4 x^2 f\,dx - \mu^2$, which reaches the original's answer $4/3$.

3. **Photoresist thickness, part (c) (Ch. 5).** The original line read "$100x + 20.50 = 0.90$"; solving $100x - 20.50 = 0.90$ gives the note's own answer $x = 0.2140$. The sign was treated as a transcription slip and the authoritative answer retained.

4. **Line-width problem $\mu = 0.4$, part (c) (Ch. 6).** The notes wrote $\frac{x-0.4}{0.05} = 1.28$ but recorded $x = 0.336$; algebraically $0.4 + 1.28(0.05) = 0.464$. Both the note's value and the algebraic value are shown so the discrepancy is transparent.

5. **Standard-normal table excerpt (Ch. 6).** The row of values for $z = 0.0$ is transcribed exactly as written in the notes (0.50000, 0.50399, 0.50398, 0.51197); the middle two are the values as they appear in the source rather than corrected textbook values.

6. **Parts-failing problem, part (c) (Ch. 7).** The original set up $D \sim \mathrm{Bin}(10, 0.0004)$ and began $\binom{10}{3}(0.0004)^3(0.9996)^7$ but did not finish the arithmetic; the value is essentially zero, which is stated.

7. **"People who become ill" part (c) (Ch. 7).** The final lines were faint. The working $\frac{(x+0.5)-200}{\sqrt{160}} = 2.33$ leading to $x \approx 230$ was reconstructed from the visible fragments ("2.33", "$229.5 + 0.5 = 230$"), matching the boxed answer 230.

8. **Poisson "asbestos" follow-up (Ch. 7).** The trailing question about $10\,\mathrm{cm}^2$ and $> 10{,}000$ particles was partly cut off at the page edge; the setup $\lambda = 10^4$ and the answer $\mathrm{P}(Z > 0) = 0.5$ are reconstructed from the visible symbols.

9. **Erlang "calls to a telephone system," part (e) (Ch. 9).** This part was faint in the scan. Using $p = \mathrm{P}(Y > 2) = 0.8754$ (as written) and $W \sim \mathrm{Bin}(10, 0.8754)$ gives the note's answer 0.2643; the intermediate $\mathrm{P}(Y > 2)$ computation was reconstructed to reach it.

10. **Gamma-function evaluation $\Gamma(9/2)$ (Ch. 9).** The note recorded $\frac{105}{16}\pi^{1/2} = 11.6317$. The factor chain $\frac{7}{2} \cdot \frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2}$ was written out to reach that value.

11. **Weibull bearing problem (Ch. 10).** The final page was very faint. The parameters $\beta = \frac{1}{2}$, $\delta = 5000$, the mean 10,000 hours, and $\mathrm{P}(X > 6000) = e^{-1.095} = 0.334$ are transcribed as the legible values; the intermediate $\Gamma(3) = 2!$ step was filled in to connect them.

12. **Diagrams.** All figures (the negative-binomial sum-of-geometrics number line, the pdf/cdf sketches, the normal and standard-normal curves, the standardizing two-axis figure, the exponential and Erlang family plots, the uniform pdf, the step-function cdf, and the hypergeometric/binomial/normal approximation block diagram) were redrawn in TikZ/pgfplots to match the hand sketches. Curve shapes are schematic representations of the originals rather than pixel-exact copies.

Everywhere not listed above, the content is a direct transcription of the notes.